

OPTIMIZATION II

MATH 466

Prof. Joseph Abeiku Ackora-Prah

Kwame Nkrumah University of Science and Technology

Department Of Mathematics



INTRODUCTION TO LINEAR PROGRAMMING

- ▶ Linear Programming Problem
- ▶ Problem Formulation
- ▶ Graphical Solution Procedure
 1. Extreme Points and the Optimal Solution
 2. Isoprofit(cost) Function Line Method
- ▶ Simplex Algorithm
- ▶ Special Cases
- ▶ Application of Linear Programming
 1. Transportation Models
 2. Transshipment Models
 3. Assignment Models



Linear Programming

- ▶ Linear programming has nothing to do with computer programming.
- ▶ The use of the word “programming” here means “choosing a course of action.”
- ▶ Linear programming involves choosing a course of action when the mathematical model of the problem contains only linear functions.



Linear Programming (LP) Problem

- ▶ The maximization or minimization of some quantity is the objective in all linear programming problems.
- ▶ All LP problems have constraints that limit the degree to which the objective can be pursued.
- ▶ A feasible solution satisfies all the problem's constraints.
- ▶ An optimal solution is a feasible solution that results in the largest possible objective function value when maximizing (or smallest when minimizing).
- ▶ A graphical solution method can be used to solve a linear program with two variables.



Feasible Solution Set and Optimal Solution

- ▶ The constraints in a linear programming problem form a system of linear inequalities, which have a solution space S .
- ▶ Each point in S is a candidate for the solution of the linear programming problem and is referred to as a feasible solution.
- ▶ The space S itself is referred to as a feasible region.
- ▶ Among all the points in the space S , the point(s) that optimizes the objective function of the linear programming problem is called an optimal solution.



Linear Programming (LP) Problem

- ▶ If both the objective function and the constraints are linear, the problem is referred to as a linear programming problem.
- ▶ Linear functions are functions in which each variable appears in a separate term raised to the first power and is multiplied by a constant (which could be 0).
- ▶ Linear constraints are linear functions that are restricted to be “less than or equal to”, “equal to”, or “greater than or equal to” a constant.



General Linear Programming problem

The general linear Programming problem (or model) with n decision variables and m constraints can be stated in the following form:

Optimize (Max. or Min.)

$$Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

Subject to the linear constraints:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n (\leq, =, \geq) b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n (\leq, =, \geq) b_2$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n (\leq, =, \geq) b_m$$

and $x_1, x_2, \dots, x_n \geq 0$



LP Problems

The above formulation can also be expressed in a compact form as follows:

Optimize (Max. or Min.)

$$Z = \sum_{j=1}^n c_j x_j \quad (\text{Objective function})$$

Subject to the linear constraints:

$$\sum_{j=1}^n a_{ij} x_j (\leq, =, \geq) b_i; \quad i = 1, 2, \dots, m \quad (\text{Constraints})$$

and:

$$x_j \geq 0; \quad j = 1, 2, \dots, n \quad (\text{Non-negativity conditions})$$



Key Terminology

- ▶ c_j 's: Coefficients representing the per-unit profit (or cost) of decision variable x_j in the objective function.
- ▶ a_{ij} 's: Technological or input-output coefficients representing the amount of resource i consumed per unit of activity x_j . These can be positive, negative, or zero.
- ▶ b_i : Total availability of the i^{th} resource. The term "resource" is used generally for any value on the right-hand side of a constraint.
- ▶ Assumption: It is assumed $b_i \geq 0$ for all i .
- ▶ Note: If $b_i < 0$, multiply both sides of constraint i by -1 to make $b_i > 0$ and reverse the inequality.



Graphing Linear Inequalities

- ▶ We've seen that a linear equation in two variables x and y

$$ax + by + c = 0$$

has a solution set that may be exhibited graphically as points on a straight line in the xy -plane.

- ▶ There is also a simple graphical representation for linear inequalities of two variables:

$$ax + by + c < 0$$

$$ax + by + c \leq 0$$

$$ax + by + c > 0$$

$$ax + by + c \geq 0$$

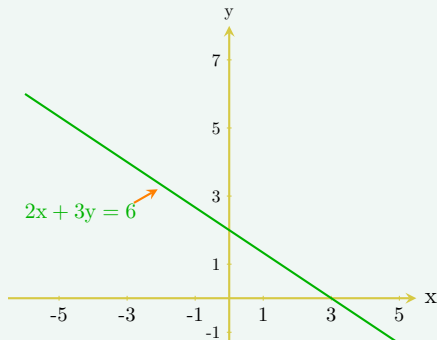


Examples

- Determine the solution set for the inequality $2x + 3y \geq 6$.

Solution

Replacing the inequality $\geq$ with an equality $=$, we obtain the equation $2x + 3y = 6$, whose graph is:

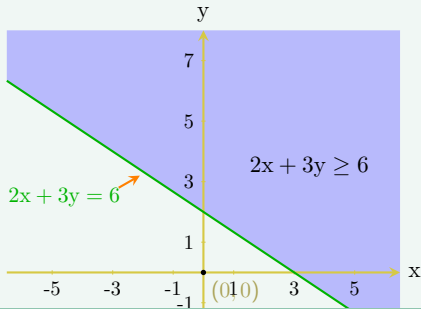


Examples

- Determine the solution set for the inequality $2x + 3y \geq 6$.

Solution

- Picking the origin as a test point, we find $2(0) + 3(0) \geq 6$, or $0 \geq 6$, which is false.
- Thus, the solution set is:



Graphing Systems of Linear Inequalities

- ▶ The solution set of a system of linear inequalities in two variables x and y is the set of all points (x, y) that satisfy each inequality of the system.
- ▶ The graphical solution of such a system may be obtained by graphing the solution set for each inequality independently and then determining the region in common with each solution set.



Example

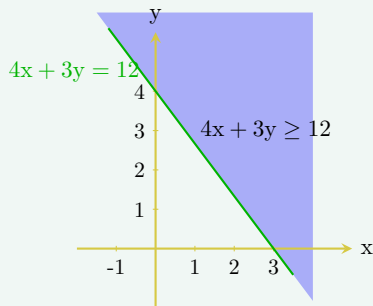
- Determine the solution set for the system

$$4x + 3y \geq 12$$

$$x - y \leq 0$$

Solution

- The **intersection** of the **solution regions** of the two inequalities represents the **solution to the system**:



Example

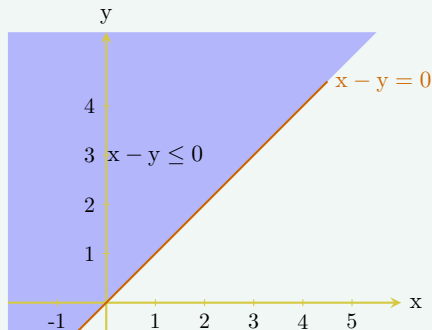
- Determine the solution set for the system

$$4x + 3y \geq 12$$

$$x - y \leq 0$$

Solution

- The **intersection** of the **solution regions** of the two inequalities represents the **solution to the system**:



Example

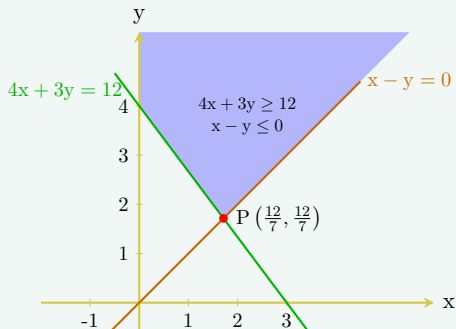
- Determine the solution set for the system

$$4x + 3y \geq 12$$

$$x - y \leq 0$$

Solution

- The **intersection** of the **solution regions** of the two inequalities represents the **solution to the system**:



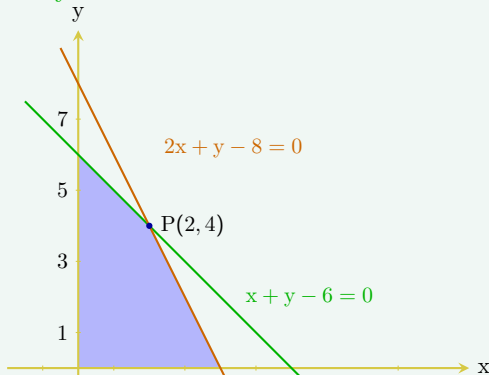
Example

Determine the solution set for the system

$$x + y - 6 \leq 0, \quad 2x + y - 8 \leq 0, \quad x \geq 0, \quad y \geq 0$$

Solution

- The **intersection** of the **solution regions** of the four inequalities represents the **solution to the system**:



Problem Formulation

- ▶ Problem formulation or modeling is the process of translating a verbal statement of a problem into a mathematical statement.
- ▶ Formulating models is an art that can only be mastered with practice and experience.
- ▶ Every LP problems has some unique features, but most problems also have common features.
- ▶ General guidelines for LP model formulation are illustrated on the slides that follow.



Guidelines on LP Model Formulation

The effective use and application requires, as a first step, the mathematical formulation of an LP model. The steps of LP model formulation are summarized as follows:

Step 1: Identify the decision variables

- (a) Express each constraint in words. For this you should first see whether the constraint is of form $\geq$ (at least, as large as), of the form $\leq$ (no larger than) or of the form $=$ (exactly, equal to).
- (b) Express verbally the objective function.
- (c) Verbally identify the decision variables with the help of Steps (a) and (b). For this you need to ask yourself the question: What decisions must be made in order to optimize the objective function?



Guidelines on LP Model Formulation cont'd...

Step 2: Identify the problem data To formulate an LP model, identify the problem data in terms of constants, and parameters associated with decision variables. It may be noted that the decision-maker can control values of the variables but cannot control values in the data set.



Guidelines on LP Model Formulation cont'd...

Step 3: Formulate the constraint Convert the verbal expression of the constraints in terms of resource requirement and availability of each resource then express each of them, as linear equality or inequality, in terms of the decision variables defined in Step 1.

Value of the decision variables in the optimal LP problem solution must satisfy these constraints in order to constitute an acceptable (feasible) solution. Wrong formulation can either lead to a solution that is not feasible or to the exclusion of a solution that is actually feasible and possibly optimal.



Guidelines on LP Model Formulation cont'd...

Step 4: Formulate the objective function

Identify whether the objective function is to be maximized or minimized, then express it in the form of a linear mathematical expression in terms of decision variables along with profit (cost) contributions associated with them.



Summarize Guidelines for Model Formulation

- ▶ Understand the problem thoroughly.
- ▶ Describe the objective.
- ▶ Describe each constraint.
- ▶ Define the decision variables.
- ▶ Write the objective in terms of the decision variables.
- ▶ Write the constraints in terms of the decision variables.



Example 1 :Model Formulation

A furniture company produces tables and chairs. Each table requires 4 hours of carpentry work and 2 hours of painting work while each chair requires 3 hours of carpentry work and 1 hour of painting work. During the current production period, 240 hours of carpentry time are available and 100 hours of painting time are available. Each table sold yields a profit of \$70 while each chair sold yields a profit of \$50. Formulate a linear programming model for this problem.



LP Formulation Table

Let x_1 and x_2 be the number of tables and chairs produced respectively.

Activity	Table (x_1)	Chair (x_2)	Availability
Carpentry (hours)	4	3	240
Painting (hours)	2	1	100
Profit (\$)	70	50	



Linear Programming Model

$$\text{Maximize: } Z = 70x_1 + 50x_2$$

Subject to:

$$4x_1 + 3x_2 \leq 240 \quad (\text{carpentry time})$$

$$2x_1 + 1x_2 \leq 100 \quad (\text{painting time})$$

$$x_1, x_2 \geq 0$$



Example 2 :Model Formulation

A paper mill produces two grades of paper namely X and Y. Owing to raw material restrictions, it cannot produce more than 400 tons of grade X and 300 tons of grade Y in a week. There are 160 production hours in a week. It requires 0.2 and 0.4 hours to produce a ton of products X and Y, respectively with corresponding profits of \$200 and \$500 per ton. Formulate the above as an LPP to maximize profit and find the optimum product mix.



LP Formulation Table

Let x_1 and x_2 be the number of units of two grades of paper of X and Y.

Resource	Grade X (x_1)	Grade Y (x_2)	Availability
Raw Material (X)	1	0	400
Raw Material (Y)	0	1	300
Production Hours	0.2	0.4	160
Profit per ton	\$200	\$500	



LP Model Formulation

Objective function: Since the profit for the two grades of paper X and Y are given, the objective function is to maximize the profit.

$$\text{Max } Z = 200x_1 + 500x_2$$

Subject to:

$$x_1 \leq 400$$

$$x_2 \leq 300$$

$$0.2x_1 + 0.4x_2 \leq 160$$

Non-negative conditions $x_1, x_2 \geq 0$.



Example 3: Model Formulation

A company manufactures two products, X and Y by using three machines A, B, and C. Machine A has 4 hours of capacity available during the coming week. Similarly, the available capacity of machines B and C during the coming week is 24 hours and 35 hours respectively. One unit of product X requires one hour of Machine A, 3 hours of machine B and 10 hours of machine C. Similarly one unit of product Y requires 1 hour, 8 hour and 7 hours of machine A, B and C respectively. When one unit of X is sold in the market, it yields a profit of \$5 per product and that of Y is \$7 per unit.



Problem Data Table

Let x_1 represents units of Product X, x_2 represents units of Product Y.

Machine	Product X (x_1)	Product Y (x_2)	Capacity (hrs)
Machine A	1	1	4
Machine B	3	8	24
Machine C	10	7	35
Profit (\$)	5	7	



Example

A manufacturing company is engaged in producing three types of products: A, B and C. The production department produces, each day, components sufficient to make 50 units of A, 25 units of B and 30 units of C. The management is confronted with the problem of optimizing the daily production of the products in the assembly department, where only 100 man-hours are available daily for assembling the products. The following additional information is available



Example on Production cont'd

Type of Product	Profit (Rs)	Assembly time (hrs)
A	12	0.8
B	20	1.7
C	45	2.5

The company has a daily commitment for 20 units of Product A and a total of 15 units of Products B and C. Formulate this problem as an LP model to maximize total profit. The model formulation data is summarized as follows:



Model Formulation

Resources/ Constraints	Product Type			Total
	A	B	C	
Production Capacity(units)	50	25	30	
Man-hours per unit	0.8	1.7	2.5	100
Other commitment(units)	20	15	(both B and C)	
Profit contribution(Rs/unit)	12	20	45	



Model Formulation

Decision variable: Let x_1, x_2 , and x_3 be the number of units of products A, B, and C to be produced, respectively.

Maximize (total profit):

$$Z = 12x_1 + 20x_2 + 45x_3$$

Subject to the constraints:

► Labour and materials:

$$0.8x_1 + 1.7x_2 + 2.5x_3 \leq 100$$

$$x_1 \leq 50$$

$$x_2 \leq 25$$

$$x_3 \leq 30$$

► Other commitment:

$$x_1 \geq 20$$

$$x_2 + x_3 \geq 15$$

and $x_1, x_2, x_3 \geq 0$



The Graphical Method

- ▶ **Optimal and Feasible Solutions:** An optimal as well as a feasible solution to an LP problem is obtained by choosing one set of values from several possible values of decision variables $x_1, x_2, \dots, x_n$ that satisfies the given constraints simultaneously and also provides an optimal (maximum or minimum) value of a given objective function.
- ▶ **Two-Variable Problems:** For an LP problem that has only two variables, it is possible that the entire set of feasible solutions can be displayed graphically by plotting linear constraints on a graph paper in order to locate the best (optimal) solution.
- ▶ **Definition:** The technique used to identify the optimal solution is called the graphical solution method (approach or technique) for an LP problem with two variables.



The Graphical Method cont'd...

Since most real-world problems have more than two decision variables, such problems cannot be solved graphically. However, the graphical approach provides understanding of solving an LP problem algebraically, involving more than two variables.

In this section, we shall discuss the following two graphical solution methods (or approaches) to find the optimal solution to an LP problem:

- ▶ Extreme point solution method
- ▶ Iso-profit (cost) function line method



Theorem 1

Linear Programming

- ▶ If a linear programming problem has a solution, then it must occur at a vertex, or corner point, of the feasible set S associated with the problem.
- ▶ If the objective function P is optimized at two adjacent vertices of S , then it is optimized at every point on the line segment joining these vertices, in which case there are infinitely many solutions to the problem.



Theorem 2

Existence of a Solution

- ▶ Suppose we are given a linear programming problem with a feasible set S and an objective function $P = ax + by$.
 - a. If S is bounded, then P has both a maximum and a minimum value on S .
 - b. If S is unbounded and both a and b are nonnegative, then P has a minimum value on S provided that the constraints defining S include the inequalities $x \geq 0$ and $y \geq 0$.
 - c. If S is the empty set, then the linear programming problem has no solution: that is, P has neither a maximum nor a minimum value.



Extreme Point Solution Methods

In this method, the coordinates of all the corner (or extreme) points of the feasible region are determined, and the value of the objective function at each of these points is computed and compared.

The coordinates of an extreme point where the optimal (maximize or minimum) value of the objective function is found represent the solution of the given LP problem.

The steps of the method are summarized as follows:



Extreme Point Solution Methods cont'd

1. Graph the feasible set.
2. Find the coordinates of all corner points (vertices) of the feasible set.
3. Evaluate the objective function at each corner point.
4. Find the vertex that renders the objective function a maximum or a minimum.
 - ▶ If there is only one such vertex, it constitutes a unique solution to the problem.
 - ▶ If there are two such adjacent vertices, there are infinitely many optimal solutions given by the points on the line segment determined by these vertices.



Procedure for Graphing Linear Inequalities

1. Draw the graph of the equation obtained for the given inequality by replacing the inequality sign with an equal sign.
 - ▶ Use a dashed or dotted line if the problem involves a strict inequality, $<$ or $>$.
 - ▶ Otherwise, use a solid line to indicate that the line itself constitutes part of the solution.
2. Pick a test point lying in one of the half-planes determined by the line sketched in step 1 and substitute the values of x and y into the given inequality.
 - ▶ Use the origin whenever possible.
3. If the inequality is satisfied, the graph of the inequality includes the half-plane containing the test point.
 - ▶ Otherwise, the solution includes the half-plane not containing the test point.



Example 1

Use the graphical method to solve the following LP problem.

Maximize: $Z = 15x_1 + 10x_2$

Subject to the constraints:

$$4x_1 + 6x_2 \leq 360$$

$$3x_1 + 0x_2 \leq 180$$

$$0x_1 + 5x_2 \leq 200$$

and $x_1, x_2 \geq 0$



Solution

To plot the constraints, we convert inequalities to equations:

1. $4x_1 + 6x_2 = 360$

▶ If $x_1 = 0$, then $6x_2 = 360 \Rightarrow x_2 = 60$. Point: $(0, 60)$

▶ If $x_2 = 0$, then $4x_1 = 360 \Rightarrow x_1 = 90$. Point: $(90, 0)$

2. $3x_1 + 0x_2 = 180$

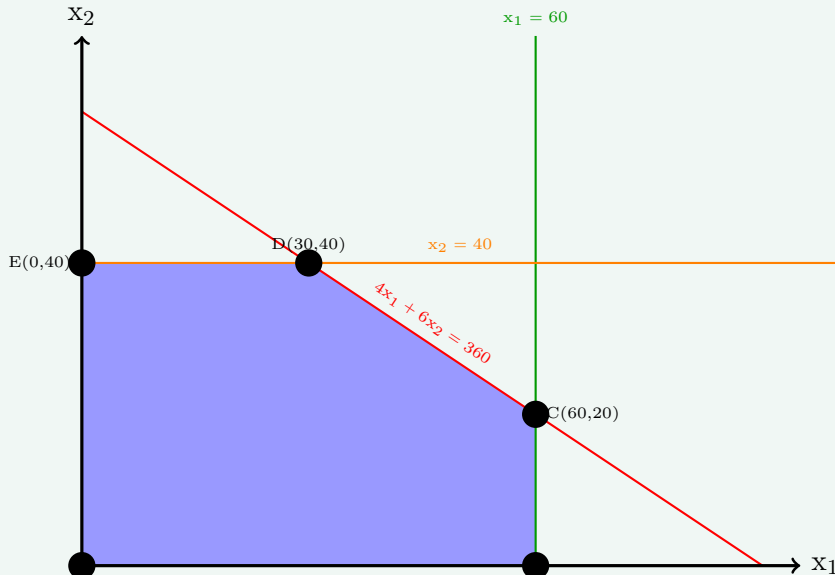
▶ This is a vertical line: $x_1 = 60$. Points: $(60, 0)$ and $(60, x_2)$

3. $0x_1 + 5x_2 = 200$

▶ This is a horizontal line: $x_2 = 40$. Points: $(0, 40)$ and $(x_1, 40)$



Solution cont'd



Calculating Total Profit ($Z = 15x_1 + 10x_2$)

Point	(x_1, x_2)	Objective Function Z
A	(0, 0)	$15(0) + 10(0) = 0$
B	(60, 0)	$15(60) + 10(0) = 900$
C	(60, 20)	$15(60) + 10(20) = 1100$
D	(30, 40)	$15(30) + 10(40) = 850$
E	(0, 40)	$15(0) + 10(40) = 400$

The Maximum value is 1100 and the optimal solution is C (60, 20).



Example 2

$$\begin{array}{ll}\text{Min} & 5x_1 + 2x_2 \\ \text{s.t.} & 2x_1 + 5x_2 \geq 10 \\ & 4x_1 - x_2 \geq 12 \\ & x_1 + x_2 \geq 4 \\ & x_1, x_2 \geq 0\end{array}$$

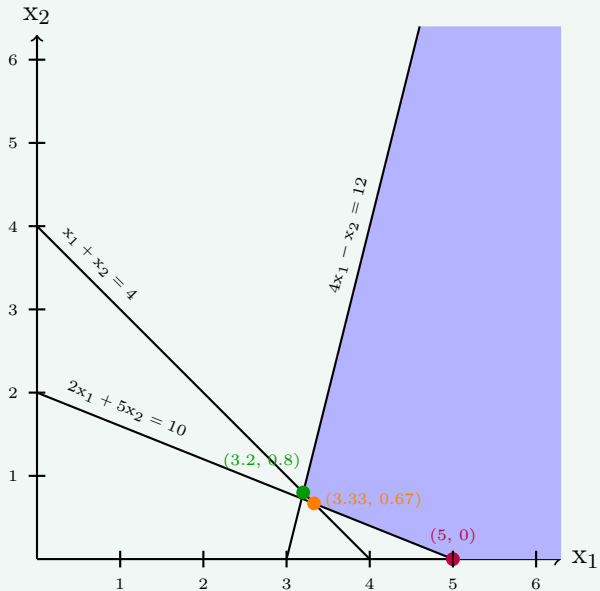


Solution

- ▶ Constraint 1: When $x_1 = 0$, then $x_2 = 2$; when $x_2 = 0$, then $x_1 = 5$.
Thus we have $(5,0)$ and $(0,2)$.
- ▶ Constraint 2: When $x_2 = 0$, then $x_1 = 3$. when x_1 to 0 will yield $x_2 = -12$.
Thus we have $(3,0)$ and $(0,-12)$.
- ▶ Constraint 3: When $x_1 = 0$, then $x_2 = 4$; when $x_2 = 0$, then $x_1 = 4$.
Thus we have $(4,0)$ and $(0,4)$.



Solution



Evaluating the Objective Function at the extreme points

Based on Theorem 1, the optimal solution occurs at an extreme point.

Point	Obj. Function: $5x_1 + 2x_2$	Value
(3.2, 0.8)	$5(3.2) + 2(0.8)$	17.6
(3.33, 0.67)	$5(3.33) + 2(0.67)$	17.99
(5, 0)	$5(5) + 2(0)$	25.0

The minimum value is 17.6 at (3.2, 0.8).



Iso-Profit (Cost) Function line Method

According to this method, the optimal solution is found by using the slope of the objective function line (equation).

An iso-profit (cost) line is a collection of points that give a solution with the same value of the objective function. By assuming various values for Z , we get different profit (cost) lines. Graphically, many such lines can be plotted parallel to each other.

The steps of the iso-profit (cost) function method are as follows:



Procedure Steps

- ▶ Step 1: Identify the feasible region and extreme points of the feasible region.
- ▶ Step 2: Draw an iso-profit (iso-cost) line for an arbitrary but small value of the objective function without violating any value of the constraints of the given problem.

However, it is simple to pick a value that gives an integer value to x_1 when setting $x_2 = 0$ and vice versa. A good choice is to use a number that is divided by the coefficients of both variables.



Procedure Steps (cont'd)

- ▶ Step 3: Move iso-profit (iso-cost) lines parallel in the direction of increasing (decreasing) objective function values.
 - ▶ The farthest line may intersect only at one corner point, providing a single optimal solution.
 - ▶ If the line coincides with a boundary, there are multiple optimal solutions along that edge.
 - ▶ If the line moves without limit, an unbounded solution exists, usually indicating a formulation error.
- ▶ Step 4: An extreme (corner) point touched by the iso-profit (or cost) line is considered the optimal solution point. The coordinates of this point yield the final value of the objective function.



Example 1: A Simple Maximization Problem

$$\text{Max} \quad 5x_1 + 7x_2$$

Objective Function

$$\text{s.t.} \quad x_1 \leq 6$$

$$2x_1 + 3x_2 \leq 19$$

$$x_1 + x_2 \leq 8$$

“Regular” Constraints

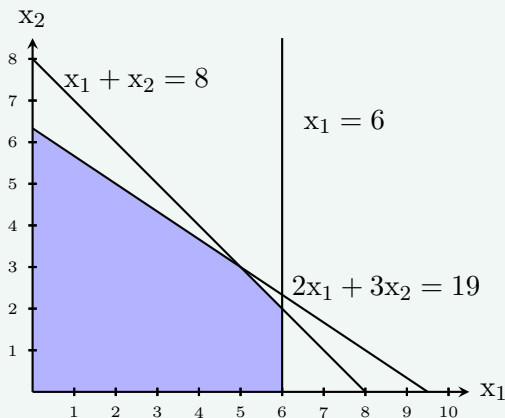
$$x_1 \geq 0 \quad \text{and} \quad x_2 \geq 0$$

Non-negativity Constraints



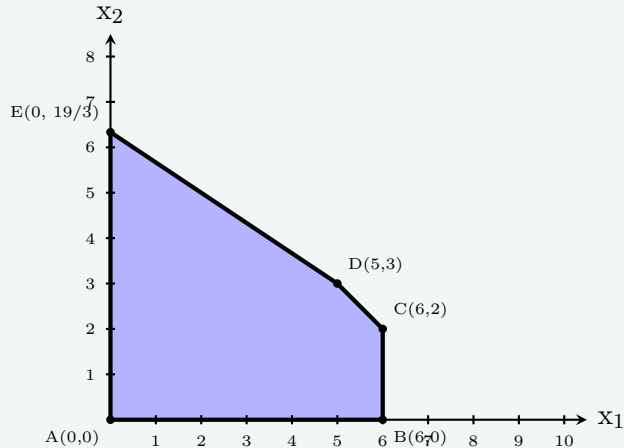
Solution

► Combined-Constraint Graph Showing Feasible Region



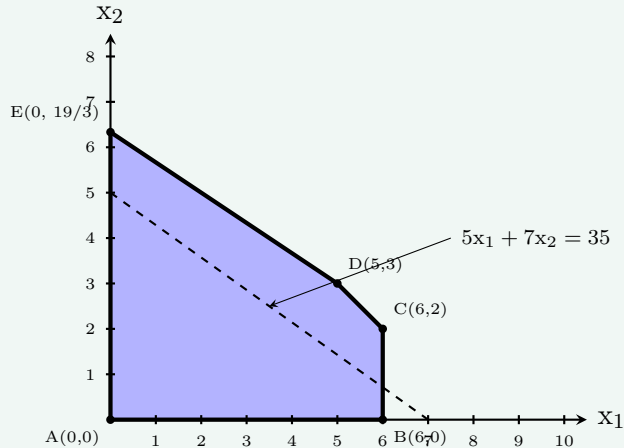
Cont'd

► Feasible Region Boundary (Perimeter)

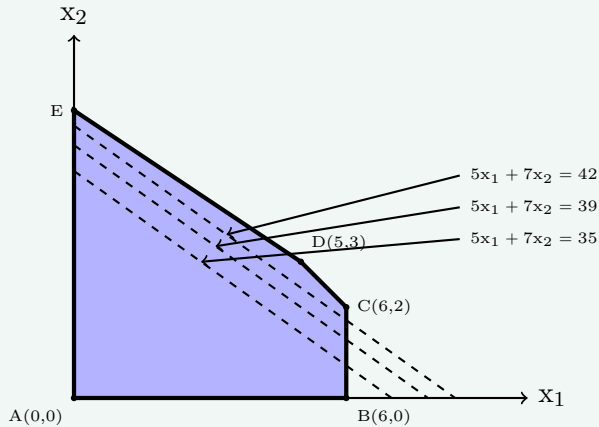


Example 1: Graphical Solution

► Feasible Region with Objective Function

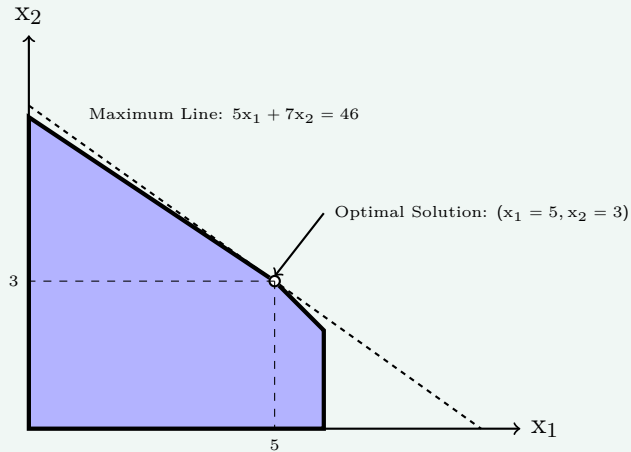


con't



cont'd

► Optimal Solution



cont'd

- ▶ Objective Function: $Z = 5x_1 + 7x_2$.
- ▶ Method:
 - ▶ Select an arbitrary value for Z (e.g., $Z = 35$) to plot the line $5x_1 + 7x_2 = 35$.
 - ▶ Shift the line parallelly in the direction of increasing Z .
- ▶ Conclusion
 - ▶ As the line moves, it passes through the feasible region.
 - ▶ The last point of contact before exiting the region is Point D(5, 3).
 - ▶ Therefore, $x_1 = 5$ and $x_2 = 3$ is the optimal solution.



Infeasibility

- ▶ No solution to the LP problem satisfies all the constraints, including the non-negativity conditions.
- ▶ Graphically, this means a feasible region does not exist.
- ▶ Causes include:
 1. A formulation error has been made.
 2. Management's expectations are too high.
 3. Too many restrictions have been placed on the problem (i.e., the problem is over-constrained).



Example: Infeasible Problem

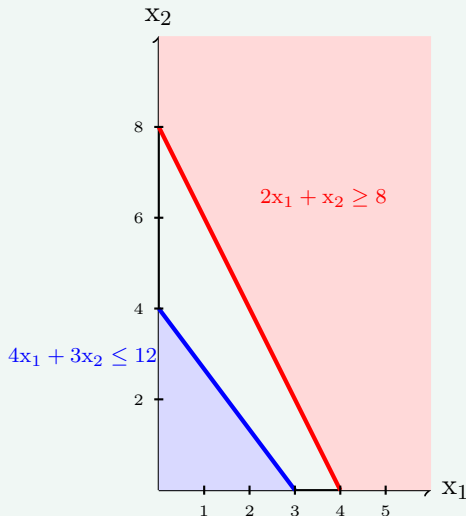
- Consider the following LP problem.

$$\begin{array}{ll}\text{Max} & 2x_1 + 6x_2 \\ \text{s.t.} & 4x_1 + 3x_2 \leq 12 \\ & 2x_1 + x_2 \geq 8 \\ & x_1, x_2 \geq 0\end{array}$$



Solution

- There are no points that satisfy both constraints, so there is no feasible region (and no feasible solution).



Unbounded Solution

- ▶ Sometimes an LP problem may have an infinite solution. Such a solution is referred as an unbounded solution. It happens when value of certain decision variables and the value of the objective function (maximization case) are permitted to increase infinitely, without violating the feasibility condition.
- ▶ For real problems, this is the result of improper formulation. (Quite likely, a constraint has been inadvertently omitted.)



Example 1: Unbounded solution

Use the graphical method to solve the following LP problem:

Maximize

$$Z = 3x_1 + 2x_2$$

Subject to the constraints:

$$x_1 - x_2 \geq 1$$

$$x_1 + x_2 \geq 3$$

$$x_1, x_2 \geq 0$$



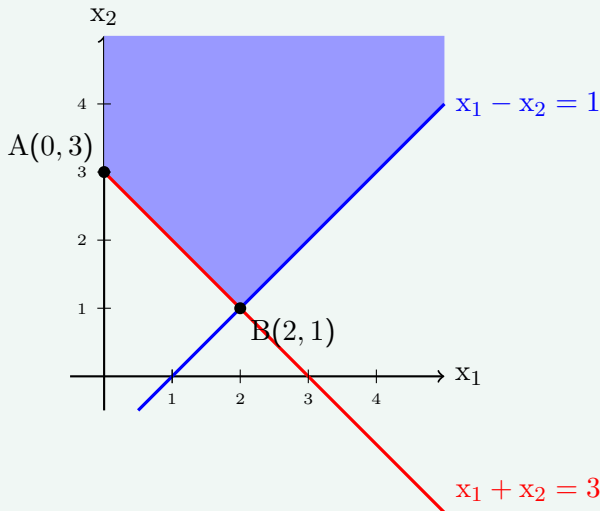
Solution: Determining Plotting Points

To graph the constraints, we treat the inequalities as equalities to find the boundary lines:

Constraint 1: $x_1 - x_2 = 1$	Constraint 2: $x_1 + x_2 = 3$
If $x_1 = 0 \implies x_2 = -1$	If $x_1 = 0 \implies x_2 = 3$
If $x_2 = 0 \implies x_1 = 1$	If $x_2 = 0 \implies x_1 = 3$
Intercepts: $(0, -1), (1, 0)$	Intercepts: $(0, 3), (3, 0)$



solution: Feasible Region



Since the given LP problem is of maximization and the feasible region is unbounded. Therefore, this linear programming problem has an unbounded solution.

Example 2 : Unbounded Solution

Use the graphical method to solve the following LP problem:

Maximize

$$Z = -4x_1 + 3x_2$$

Subject to the constraints:

$$x_1 - x_2 \leq 0$$

$$x_1 \leq 4$$

$$x_1, x_2 \geq 0$$



Solution

To graph the constraints, we convert the inequalities into equations to find boundary lines:

For the first Constraint : $x_1 - x_2 \leq 0$

- ▶ Boundary equation: $x_1 = x_2$
- ▶ Points: $(0, 0), (2, 2), (4, 4)$

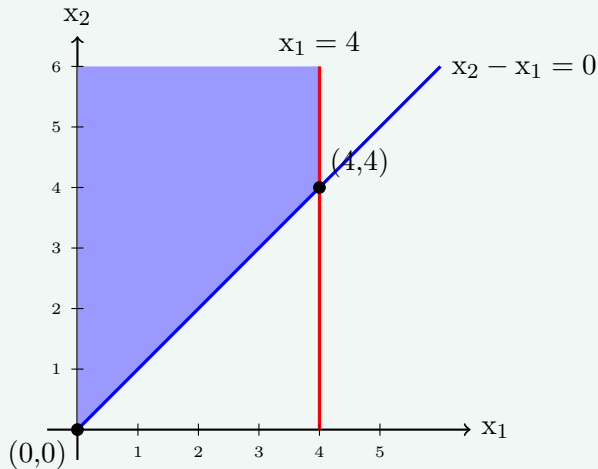
For the second constraint: $x_1 \leq 4$

- ▶ Boundary equation: $x_1 = 4$
- ▶ This is a vertical line passing through $x_1 = 4$ for all values of x_2 .

Non-negativity: $x_1, x_2 \geq 0$ (The feasible region is restricted to the first quadrant).



Solution: Feasible Region Visualization



Since objective function is of maximization, therefore, the value of Z can be made arbitrarily large. Hence, this LP problem has an unbounded solution.



Applied Example 1: A Production Problem

- ▶ Ace Novelty wishes to produce two types of souvenirs: type-A will result in a profit of \$1.00, and type-B in a profit of \$1.20.
- ▶ To manufacture a type-A souvenir requires 2 minutes on machine I and 1 minute on machine II.
- ▶ A type-B souvenir requires 1 minute on machine I and 3 minutes on machine II.
- ▶ There are 3 hours available on machine I and 5 hours available on machine II.
- ▶ How many souvenirs of each type should Ace make in order to maximize its profit?



Solution

Let's first tabulate the given information:

	Type-A	Type-B	Time Available
Profit/Unit	\$1.00	\$1.20	
Machine I	2 min	1 min	180 min
Machine II	1 min	3 min	300 min

Let x be the number of type-A souvenirs and y the number of type-B souvenirs to be made.



cont'd

	Type-A	Type-B	Time Available
▶ Profit/Unit	\$1.00	\$1.20	
Machine I	2 min	1 min	180 min
Machine II	1 min	3 min	300 min

- ▶ Then, the total profit (in dollars) is given by:

$$P = x + 1.2y$$

which is the objective function to be maximized.



cont'd

- ▶ Let's first tabulate the given information:

	Type-A	Type-B	Time Available
Profit/Unit	\$1.00	\$1.20	
Machine I	2 min	1 min	180 min
Machine II	1 min	3 min	300 min

- ▶ The total amount of time that machine I is used is:

$$2x + y$$

and must not exceed 180 minutes.

- ▶ Thus, we have the inequality:

$$2x + y \leq 180$$



con't

- ▶ Let's first tabulate the given information:

	Type-A	Type-B	Time Available
Profit/Unit	\$1.00	\$1.20	
Machine I	2 min	1 min	180 min
Machine II	1 min	3 min	300 min

- ▶ The total amount of time that machine II is used is:

$$x + 3y$$

and must not exceed 300 minutes.

- ▶ Thus, we have the inequality:

$$x + 3y \leq 300$$



con't

- ▶ Let's first tabulate the given information:

	Type-A	Type-B	Time Available
Profit/Unit	\$1.00	\$1.20	
Machine I	2 min	1 min	180 min
Machine II	1 min	3 min	300 min

- ▶ Finally, neither x nor y can be negative, so:

$$x \geq 0$$

$$y \geq 0$$



Applied Example 1: Solution

- ▶ In short, we want to maximize the objective function:

$$P = x + 1.2y$$

- ▶ subject to the system of inequalities:

$$2x + y \leq 180$$

$$x + 3y \leq 300$$

$$x \geq 0$$

$$y \geq 0$$

- ▶ We can now solve the problem graphically.

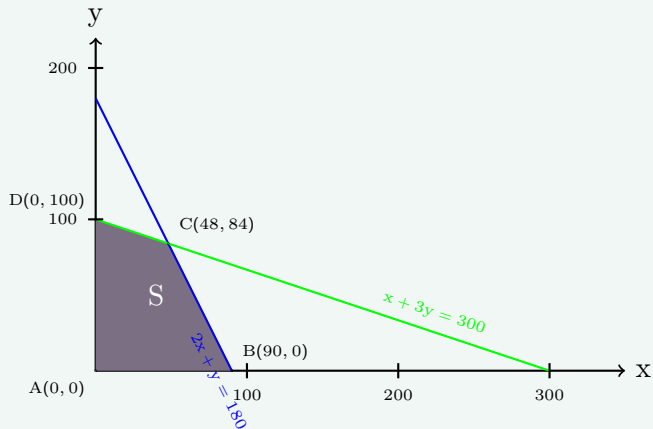


cont'd

- ▶ Constraint 1: $2x + y \leq 180$
 - ▶ When $x = 0$, $y = 180$. Point: $(0, 180)$
 - ▶ When $y = 0$, $2x = 180 \implies x = 90$. Point: $(90, 0)$
- ▶ Constraint 2: $x + 3y \leq 300$
 - ▶ When $x = 0$, $3y = 300 \implies y = 100$. Point: $(0, 100)$
 - ▶ When $y = 0$, $x = 300$. Point: $(300, 0)$
- ▶ Non-negativity:
 - ▶ $x \geq 0, y \geq 0$. Point: $(0, 0)$



Graphical Solution: Feasible Region



Evaluating the Objective Function $P = x + 1.2y$

To maximize profit, we test each vertex of the feasible region S:

Vertex	Calculation: $x + 1.2y$	Total Profit (P)
A(0, 0)	$0 + 1.2(0)$	\$0
B(90, 0)	$90 + 1.2(0)$	\$90
C(48, 84)	$48 + 1.2(84)$	$48 + 100.8 = \$148.8$
D(0, 100)	$0 + 1.2(100)$	\$120

The maximum profit is \$148.8, achieved by producing 48 Type-A and 84 Type-B souvenirs.



Applied Example 2: A Nutrition Problem

- ▶ A nutritionist advises an individual who is suffering from iron and vitamin B deficiency to take at least 2400 milligrams (mg) of iron, 2100 mg of vitamin B₁, and 1500 mg of vitamin B₂ over a period of time.
- ▶ Two vitamin pills are suitable, brand-A and brand-B.
- ▶ Each brand-A pill costs 6 cents and contains 40 mg of iron, 10 mg of vitamin B₁, and 5 mg of vitamin B₂.
- ▶ Each brand-B pill costs 8 cents and contains 10 mg of iron and 15 mg each of vitamins B₁ and B₂.
- ▶ What combination of pills should the individual purchase in order to meet the minimum iron and vitamin requirements at the lowest cost?



Solution

To solve this, we first organize the information:

	Brand-A	Brand-B	Min. Requirement
Cost/Pill	6¢	8¢	-
Iron	40 mg	10 mg	2400 mg
Vitamin B ₁	10 mg	15 mg	2100 mg
Vitamin B ₂	5 mg	15 mg	1500 mg

Let x be the number of brand-A pills and y be the number of brand-B pills to be purchased.



- ▶ Objective Function: The cost C (in cents) is given by:

$$C = 6x + 8y$$

(This is the objective function to be minimized.)

- ▶ Iron Constraint: The amount of iron is given by $40x + 10y$ mg, and this must be ≥ 2400 mg:

$$40x + 10y \geq 2400$$

- ▶ Vitamin B_1 Constraint: The amount of vitamin B_1 is given by $10x + 15y$ mg, and this must be ≥ 2100 mg:

$$10x + 15y \geq 2100$$

- ▶ Vitamin B_2 Constraint: The amount of vitamin B_2 is given by $5x + 15y$ mg, and this must be ≥ 1500 mg:

$$5x + 15y \geq 1500$$



Solution

We want to minimize the objective function:

$$C = 6x + 8y$$

Subject to the system of inequalities:

$$40x + 10y \geq 2400$$

$$10x + 15y \geq 2100$$

$$5x + 15y \geq 1500$$

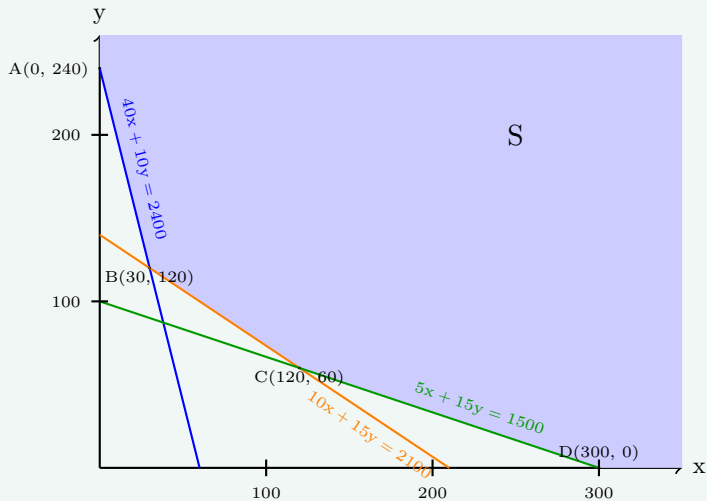
$$x \geq 0$$

$$y \geq 0$$

We can now solve the problem graphically.



Constraint Plot



cont'd

The objective function to minimize is $C = 6x + 8y$. We evaluate this at the vertices of the feasible region S:

Vertex (x, y)	Calculation	Cost C (cents)
A(0, 240)	$6(0) + 8(240)$	1920
B(30, 120)	$6(30) + 8(120)$	1140
C(120, 60)	$6(120) + 8(60)$	1200
D(300, 0)	$6(300) + 8(0)$	1800

- ▶ We can see that C is minimized at the vertex B(30, 120) and has a value of 1140
- ▶ Recalling what the symbols x, y, and C represent, we conclude that the individual should purchase 30 brand-A pills and 120 brand-B pills at a minimum cost of \$11.40.



Important Definitions

Optimal Solution: The set of values for decision variables $x_j (j = 1, 2, \dots, n)$ that satisfy the constraints of an LP problem.

Feasible Solution: The set of values for decision variables $x_j (j = 1, 2, \dots, n)$ that satisfy all constraints and non-negativity conditions simultaneously.

Infeasible Solution: The set of values for decision variables $x_j (j = 1, 2, \dots, n)$ that do not satisfy all constraints and non-negativity conditions simultaneously.



Important Definitions cont'd

Basic Solution: For m simultaneous equations in n variables ($n > m$), a solution obtained by setting $(n - m)$ variables to zero and solving the remaining m equations.

- ▶ Non-basic variables: The $(n - m)$ variables set to zero.
- ▶ Basic variables: The remaining m variables.

Basic Feasible Solution : A basic solution where all basic variables assume non-negative values. The Basic Feasible Solution is categorized into two types.

- ▶ Degenerate: A basic feasible solution is degenerate if the value of at least one basic variable is zero.
- ▶ Non-degenerate: A basic feasible solution is non-degenerate if the values of all m basic variables are strictly positive.



Important Definitions (cont'd)

Optimum Basic Feasible Solution: A basic feasible solution that optimizes (maximizes or minimizes) the objective function value.

Unbounded Solution: A solution where the objective function value can increase or decrease infinitely within the feasible region.



The Simplex Method

- ▶ The Simplex method is a computational routine of repeating a procedure until an optimal solution is attained.
- ▶ It is known as an iterative procedure.
- ▶ The method indicates when the optimal solution has been reached and thus the iteration can be terminated.
- ▶ Its main aim is to determine the solution that optimizes the objective function from the infinite solutions within the feasible region.



SIMPLEX METHOD

- ▶ The first step in the simplex method for solving the general LP problem is to convert the system of m linear inequalities equation (2) to a system of m linear equations.

$$Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n \quad (1)$$

subject to

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &\leq b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &\leq b_2 \\ &\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &\leq b_m \end{aligned} \quad (2)$$



SIMPLEX METHOD

- ▶ This is done by including “slack variables” $s_1, s_2, \dots, s_m$, one for each inequality, so that the system becomes:

$$\begin{array}{rccccccc} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n & + s_1 & & & & & = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n & & + s_2 & & & & = b_2 \\ \vdots & \vdots & & \vdots & & \ddots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n & & & & + s_m & & = b_m \end{array} \quad (3)$$

The objective function becomes:

$$Z = c_1x_1 + c_2x_2 + \dots + c_nx_n + 0s_1 + 0s_2 + \dots + 0s_m$$



Standard LP Form

The LP problem can be restated as follows:

$$\text{Maximize } Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n + 0s_1 + 0s_2 + \cdots + 0s_m$$

Subject to:

$$\begin{array}{ccccccc} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n + s_1 & & & & & & = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & +s_2 & & & & & = b_2 \\ \vdots & \vdots & & \vdots & & \ddots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & & & & +s_m & & = b_m \end{array} \quad (4)$$



Role of Slack Variables

- ▶ The slack variables are merely a device used in arriving at the optimal solution and may or may not have physical meanings in individual problems.
- ▶ If we knew which coordinates were zero at the optimum, we could insert their values into the objective function and compute for the optimum.
- ▶ If not, then a step by step method can be used until the optimal solution is obtained.



Example 1

Let x_1 and x_2 be the number of chairs and tables to be produced by the manufacturer using available resources of timber and labor.

Maximize

$$Z = 6x_1 + 8x_2$$

subject to

$$30x_1 + 20x_2 \leq 300$$

$$5x_1 + 10x_2 \leq 110$$

$$x_1 \geq 0, \quad x_2 \geq 0$$



Solution

Introducing the slack variables s_1 and s_2 , the problem can be restated as:

Maximize

$$Z = 6x_1 + 8x_2 + 0s_1 + 0s_2$$

subject to

$$30x_1 + 20x_2 + s_1 = 300$$

$$5x_1 + 10x_2 + s_2 = 110$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Where x_1 and x_2 are the decision variables and s_1 and s_2 are slack variables. It can be simplified in a tabular form as below:



SIMPLEX TABLEAU I:

INITIAL SIMPLEX TABLEAU

C_B	Basis	x_1	x_2	s_1	s_2	RHS
0	s_1	30	20	1	0	300
0	s_2	5	10	0	1	110
	C_j	6	8	0	0	
	Z_j					
	$C_j - Z_j$					



SIMPLEX TABLEAU II

Iterative Step

C_B	Basis	x_1	x_2	s_1	s_2	RHS	θ
0	s_1	30	20	1	0	300	15
0	s_2	5	10	0	1	110	11 $\rightarrow$
	C_j	6	8	0	0		
	Z_j	0	0	0	0	0	
	$C_j - Z_j$	6	8 $\uparrow$	0	0		



SIMPLEX TABLEAU III

Iterative Step: Next Tableau

C_B	Basis	x_1	x_2	s_1	s_2	RHS	θ
0	s_1	20	0	1	-2	80	$4 \rightarrow$
8	x_2	$1/2$	1	0	$1/10$	11	22
	C_j	6	8	0	0		
	Z_j	4	8	0	$8/10$	88	
	$C_j - Z_j$	$2\uparrow$	0	0	$-8/10$		



SIMPLEX TABLEAU IV: FINAL STEP

C_B	Basis	x_1	x_2	s_1	s_2	RHS
6	x_1	1	0	$1/20$	$-1/10$	4
8	x_2	0	1	$-1/40$	$3/20$	9
	C_j	6	8	0	0	
	Z_j	6	8	$1/10$	$6/10$	96
	$C_j - Z_j$	0	0	$-1/10$	$-6/10$	

- An optimal solution is reached because all values in the $C_j - Z_j$ row are ≤ 0 . Thus $x_1 = 4$ and $x_2 = 9$

Objective function = $6x_1 + 8x_2 = 6(4) + 8(9) = 96$

∴ The optimal solution is $x_1 = 4$, $x_2 = 9$ and the optimal(maximum) value is 96



Example 2

Minimize

$$Z = x_1 - 3x_2 + 3x_3$$

Subject to

$$3x_1 - x_2 + 2x_3 \leq 7$$

$$2x_1 + 4x_2 \geq -12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

$$x_1, x_2, x_3 \geq 0$$



Solution

Minimize

$$Z = x_1 - 3x_2 + 3x_3$$

Subject to

$$3x_1 - x_2 + 2x_3 \leq 7$$

$$-2x_1 - 4x_2 \leq 12 \quad (\times -1 \text{ to ensure } b_i \geq 0)$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

$$x_1, x_2, x_3 \geq 0$$

Note: We multiply the second constraint by -1 to ensure the RHS (b value) is positive, resulting in $2x_1 + 4x_2 \geq -12$.



Standard Form

Converting to Standard Form

Minimize

$$Z = x_1 - 3x_2 + 3x_3 + 0s_1 + 0s_2 + 0s_3$$

Subject to

$$3x_1 - x_2 + 2x_3 + s_1 = 7$$

$$-2x_1 - 4x_2 + s_2 = 12$$

$$-4x_1 + 3x_2 + 8x_3 + s_3 = 10$$

Non-negativity Constraints:

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$



Simplex Tableau

Simplex Tableau

C_i	Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS
0	s_1	3	-1	2	1	0	0	7
0	s_2	-2	-4	0	0	1	0	12
0	s_3	-4	3	8	0	0	1	10
	C_j	1	-3	3	0	0	0	
	Z_j							
	$C_j - Z_j$							



Simplex Tableau I: Initial Step

Initial Simplex Tableau

C_i	Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS	θ_i
0	s_1	3	-1	2	1	0	0	7	-7
0	s_2	-2	-4	0	0	1	0	12	-3
0	s_3	-4	3	8	0	0	1	10	$10/3 \rightarrow$
	C_j	1	-3	3	0	0	0		
	Z_j	0	0	0	0	0	0	0	
	$C_j - Z_j$	1	$-3\uparrow$	3	0	0	0		

The initial feasible solution is:

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 0, \quad s_1 = 7, \quad s_2 = 12, \quad s_3 = 10$$



Simplex Tableau I: Initial Step

Initial Simplex Tableau

C_i	Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS	θ_i
0	s_1	3	-1	2	1	0	0	7	-7
0	s_2	-2	-4	0	0	1	0	12	-3
0	s_3	-4	3	8	0	0	1	10	$10/3 \rightarrow$
	C_j	1	-3	3	0	0	0		
	Z_j	0	0	0	0	0	0	0	
	$C_j - Z_j$	1	-3 $\uparrow$	3	0	0	0		

The initial feasible solution is:

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 0, \quad s_1 = 7, \quad s_2 = 12, \quad s_3 = 10$$



SIMPLEX TABLEAU II

Iterative Step: Simplex Tableau II

C_i	Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS	θ_i
0	s_1	$5/3$	0	$14/3$	1	0	$1/3$	$31/3$	$31/5 \rightarrow$
0	s_2	$-22/3$	0	$32/3$	0	1	$4/3$	$76/3$	$-76/22$
-3	x_2	$-4/3$	1	$8/3$	0	0	$1/3$	$10/3$	$-10/4$
	C_j	1	-3	3	0	0	0		
	Z_j	4	-3	-8	0	0	0	-10	
	$C_j - Z_j$	$-3\uparrow$	0	11	0	0	0		



SIMPLEX TABLEAU III: FINAL STEP

C_i	Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS	θ_i
1	x_1	1	0	$14/5$	$3/5$	0	$1/5$	$31/5$	
0	s_2	0	0	$156/5$	$22/5$	1	$14/5$	$354/5$	
-3	x_2	0	1	$32/5$	$4/5$	0	$3/5$	$56/5$	
	C_j	1	-3	3	0	0	0		
	Z_j	1	-3	-16.4	-1.8	0	-1.6	-28.6	
	$C_j - Z_j$	0	0	19.4	1.8	0	1.6		

- ▶ In a minimization problem, optimality is reached when all values in the $C_j - Z_j$ row are ≥ 0 . This indicates that no further reduction in Z is possible.
- ▶ Optimal Solution: $x_1 = \frac{31}{5}$, $x_2 = \frac{56}{5}$, $x_3 = 0$.



Example 3

Maximize

$$Z = 5x_1 - 3x_2$$

Subject to

$$x_1 + x_2 \leq 2$$

$$5x_1 + 2x_2 \leq 10$$

$$3x_1 + 8x_2 \leq 12$$

$$x_1, x_2 \geq 0$$



Converting to Standard Form

Maximize

$$Z = 5x_1 - 3x_2 + 0s_1 + 0s_2 + 0s_3$$

Subject to

$$x_1 + x_2 + s_1 = 2$$

$$5x_1 + 2x_2 + s_2 = 10$$

$$3x_1 + 8x_2 + s_3 = 12$$

Non-negativity Constraints:



Simplex Tableau I

Initial Simplex Tableau

C_i	Basis	x_1	x_2	s_1	s_2	s_3	RHS	θ_i
0	s_1	1	1	1	0	0	2	$2 \rightarrow$
0	s_2	5	2	0	1	0	10	2
0	s_3	3	8	0	0	1	12	4
	C_j	5	3	0	0	0		
	Z_j	0	0	0	0	0	0	
	$C_j - Z_j$	$5\uparrow$	3	0	0	0		



Optimal Solution

C_i	Basis	x_1	x_2	s_1	s_2	s_3	RHS
5	x_1	1	1	1	0	0	2
0	s_2	0	-3	-5	1	0	0
0	s_3	0	5	-3	0	0	6
	C_j	5	3	0	0	0	
	Z_j	5	5	5	0	0	10
	$C_j - Z_j$	0	-2	-5	0	0	

- ▶ In this maximization problem, the optimal solution is reached because all values in the $C_j - Z_j$ row are ≤ 0 . This means no further increase in Z is possible.
- ▶ Optimal Solution: $x_1 = 2$, $x_2 = 0$.
- ▶ Maximum Value $Z = 5(2) - 3(0) = 10$.



Tie for Entering Basic Variable (Key Column)

While solving an LP problem using simplex method two or more columns of simplex table may have same $c_j - z_j$ value (positive or negative depending upon the type of LP problem). In order to break this tie, the selection for key column (entering variable) can be made arbitrary. However, the number of iterations required to arrive at the optima solution can be minimized by adopting the following rules:



Tie for Entering Basic Variable (Key Column)

While solving an LP problem using simplex method two or more columns of simplex table may have same $c_j - z_j$ value (positive or negative depending upon the type of LP problem). In order to break this tie, the selection for key column (entering variable) can be made arbitrary. However, the number of iterations required to arrive at the optima solution can be minimized by adopting the following rules:

1. If there is a tie between two decision variables, then the selection can be made arbitrarily.



Tie for Entering Basic Variable (Key Column)

While solving an LP problem using simplex method two or more columns of simplex table may have same $c_j - z_j$ value (positive or negative depending upon the type of LP problem). In order to break this tie, the selection for key column (entering variable) can be made arbitrary. However, the number of iterations required to arrive at the optima solution can be minimized by adopting the following rules:

1. If there is a tie between two decision variables, then the selection can be made arbitrarily.
2. If there is a tie between a decision variable and a slack (or surplus) variable, then select the decision variable to enter into basis.



Tie for Entering Basic Variable (Key Column)

While solving an LP problem using simplex method two or more columns of simplex table may have same $c_j - z_j$ value (positive or negative depending upon the type of LP problem). In order to break this tie, the selection for key column (entering variable) can be made arbitrary. However, the number of iterations required to arrive at the optima solution can be minimized by adopting the following rules:

1. If there is a tie between two decision variables, then the selection can be made arbitrarily.
2. If there is a tie between a decision variable and a slack (or surplus) variable, then select the decision variable to enter into basis.
3. If there is a tie between two slack (or surplus) variables, then the selection can be made arbitrarily.



Tie for Leaving Basic Variable (Key Row) - Degeneracy

- ▶ While solving an LP problem a situation may arise where either the minimum ratio to identify the basic variable to leave the basis is not unique or value of one or more basic variable in the x_B becomes zero. This causes the problem of degeneracy.
- ▶ Usually, the problem of degeneracy arises due to redundant constraint, i.e. one or more of the constraints of LP problem are not part of feasible solution space.
- ▶ For example, constraints such as $x_1 \leq 5$, $x_2 \leq 5$ and $x_1 + x_2 \leq 5$ in the LP problem make constraint $x_1 + x_2 \leq 5$ unnecessary (redundant).



Cont'd...

Degeneracy may occur at any iteration of the simplex method. In order to break the tie in the minimum ratios, the selection can be made arbitrarily. However, the number of iterations required to arrive at the optima solution can be minimized by adopting the following rules:

1. Divide the coefficients of slack variables in the simplex table where degeneracy is seen by the corresponding positive numbers of the key column in the row, starting from left to right.
2. Compare the ratios in step (i) from left to right column-wise, select the row that contains the smallest ratio.

When there is a tie between a slack and artificial variable to leave the basis, preference should be given to the artificial variable for leaving the basis.



Example

Solve the following LP problem:

Maximize

$$Z = 3x_1 + 9x_2$$

Subject to the constraints:

$$x_1 + 4x_2 \leq 8$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$



Writing it in the Standard Form

Maximize

$$Z = 3x_1 + 9x_2 + 0s_1 + 0s_2$$

Subject to

$$x_1 + 4x_2 + s_1 = 8$$

$$x_1 + 2x_2 + s_2 = 4$$

Non-negativity Constraints:

$$x_1, x_2, s_1, s_2 \geq 0$$



SIMPLEX TABLEAU

INITIAL SIMPLEX TABLEAU

C_B	Basis	x_1	x_2	s_1	s_2	RHS
0	s_1	1	4	1	0	8
0	s_2	1	2	0	1	4
	C_j	0	0	0	0	
	Z_j					
	$C_j - Z_j$					



SIMPLEX TABLEAU I:

INITIAL SIMPLEX TABLEAU

C_B	Basis	x_1	x_2	s_1	s_2	RHS	θ
0	s_1	1	4	1	0	8	$8/4 = 2$
0	s_2	1	2	0	1	4	$4/2 = 2$
	C_j	3	9	0	0		
	Z_j	0	0	0	0	0	
	$C_j - Z_j$	3	9	0	0		

↑



Solution cont'd

Row	x_2 -column (Key Column)	Column	
		s_1	s_2
s_1	4	1	0
s_2	2	0	1



Solution Contd...

- (ii) Dividing the coefficients of slack variables s_1 and s_2 by the corresponding elements in the key column as shown below in the table:

Row	x ₂ -column (Key Column)	<u>Column</u>	
		s ₁	s ₂
s ₁	4	$\frac{1}{4} = \frac{1}{4}$	$\frac{0}{4} = 0$
s ₂	2	$\frac{0}{2} = 0$	$\frac{1}{2} = \frac{1}{2}$

- (iii) Comparing the ratios of Step (ii) from left to right column-wise, the minimum ratio (i.e., $0/2 = 0$) occurs in the s_2 -row. Thus, variable s_2 is selected to leave the basis. The new solution is shown in the Table in the next slides.



cont'd

C_B	Basis	x_1	x_2	s_1	s_2	RHS	θ
0	s_1	1	4	1	0	8	2
0	s_2	1	②	0	1	4	$2 \rightarrow$
	C_j	3	9	0	0		
	Z_j	0	0	0	0	0	
	$C_j - Z_j$	3	9	0	0		

↑



cont'd

C_B	Basis	x_1	x_2	s_1	s_2	RHS
0	s_1	-1	0	1	-2	0
9	x_2	1/2	1	0	1/2	2
	Z_j	9/2	9	0	9/2	18
	$C_j - Z_j$	-3/2	0	0	-9/2	

Since all $C_j - Z_j \leq 0$, the optimal solution is reached. The optimal basic solution is $x_1 = 0$, $x_2 = 2$ and $\text{Max } Z = 18$



Simplex Algorithm: The ($\geq$ and $=$) Case

When the constraints are of the $\geq$ type:

$$\sum_{j=1}^n a_{ij}x_j \geq b_i, \quad x_j \geq 0$$

To convert this into an equation, we subtract a surplus (negative slack) variable s_i :

$$\sum_{j=1}^n a_{ij}x_j - s_i = b_i, \quad x_j, s_i \geq 0$$

This solution is not feasible because it does not satisfy non-negativity conditions of surplus variables (i.e. $s_i \geq 0$)



Artificial Variables

To obtain an initial feasible solution, we add artificial variables a_i :

$$\sum_{j=1}^n c_{ij}x_j - s_i + a_i = b_i$$

Key Concepts:

- ▶ The system now has $(n + 2m)$ variables.
- ▶ Setting x_j and s_i to zero gives an initial solution $a_i = b_i$, which is feasible.
- ▶ Since a_i are not part of the original problem, they must be removed from the optimal solution.



Methods to remove artificial variables:

1. Big-M Method (Method of Penalties): Assigns a very large cost (M) to artificial variables to force them to zero.
2. Two-Phase Method: Eliminates artificial variables in Phase I.



Remark: The Role of Artificial Variables

- ▶ Artificial variables have no physical meaning; they are purely mathematical tools used to generate an initial basic feasible solution.
- ▶ Lifecycle: They must be removed (or "dropped out") from the solution mix before the optimal solution is reached.
- ▶ Implementation: This removal is achieved by assigning appropriate penalty coefficients to these variables within the objective function.
- ▶ Usage: Artificial variables are exclusively added to constraints that contain either:
 - ▶ An equality (=) sign.
 - ▶ A greater than or equal to ($\geq$) sign.



The Big-M Method

The Big-M method removes artificial variables by assigning a very large "penalty" coefficient to them in the objective function:

- ▶ Minimization: Assign $+M$ (a large positive penalty).
- ▶ Maximization: Assign $-M$ (a large negative penalty).
- ▶ Here, $M > 0$ represents an arbitrarily large number.

Iterative Procedure

1. Standard Form: Add slack/surplus and artificial variables.
 - ▶ Slack/Surplus variables get a coefficient of 0.
 - ▶ Artificial variables get $+M$ (min) or $-M$ (max) coefficients.
2. Initial Solution: Obtained by setting all decision variables to zero, leaving artificial variables as the initial basic variables.



Big-M Method: Iterative Procedure cont'd

- 3 Calculate $c_j - z_j$: Evaluate the bottom row of the simplex table.
 - ▶ If all $c_j - z_j \geq 0$, the current solution is optimal.
 - ▶ If any $c_j - z_j < 0$ (for minimization), the variable with the largest negative value enters the basis.
 - ▶ If a column has negative $c_j - z_j$ but no positive entries, the solution is unbounded.
- 4 Pivot Selection: Determine the key row and key element (pivot) using the same ratio rules as the standard Simplex algorithm.
- 5 Iteration: Repeat the process of calculating ratios and pivoting until an optimal solution is reached (all artificial variables are driven to zero).



Example : Big-M Method

Use the penalty (Big-M) method to solve the following LP problem:

Minimize:

$$Z = 5x_1 + 3x_2$$

Subject to the constraints:

$$2x_1 + 4x_2 \leq 12$$

$$2x_1 + 2x_2 = 10$$

$$5x_1 + 2x_2 \geq 10$$

and $x_1, x_2 \geq 0$



Solution

To simplify the construction of the Simplex table, we align all variables, leaving empty spaces where coefficients are zero.

Standard Form

$$2x_1 + 4x_2 + s_1 \qquad \qquad = 12$$

$$2x_1 + 2x_2 \qquad \qquad + a_2 \qquad = 10$$

$$5x_1 + 2x_2 \qquad - s_3 \qquad + a_3 = 10$$

and

$$x_1, x_2, s_1, s_3, a_2, a_3 \geq 0$$

Minimize

$$Z = 5x_1 + 3x_2 + 0s_1 + 0s_3 + Ma_2 + Ma_3$$



Solution

		Variables						
c_i	Basis	x_1	x_2	s_1	s_3	a_2	a_3	RHS
0	s_1	2	4	1	0	0	0	12
M	a_2	2	2	0	0	1	0	10
M	a_3	5	2	0	-1	0	1	10
	C_j	5	3	0	0	M	M	
	z_j							
	$c_j - z_j$							



SIMPLEX TABLEAU II

c_B	Basis	Variables						RHS	θ_i
		x_1	x_2	s_1	s_3	a_2	a_3		
0	s_1	2	4	1	0	0	0	12	$12/2 = 6$
M	a_2	2	2	0	0	1	0	10	$10/2 = 5$
M	a_3	⑤	2	0	-1	0	1	10	$10/5 = 2 \rightarrow$
	C_j	5	3	0	0	M	M	$Z = 20M$	
	z_j	7M	4M	0	-M	M	M		
	$c_j - z_j$	$5 - 7M$	$3 - 4M$	0	M	0	0		

↑

The Initial Feasible Solution is $x_1 = 0, x_2 = 0, s_3 = 0, s_1 = 12, a_2 = 10, a_3 = 10$.



Simplex Tableau II: Second Iteration

c_B	Basis	x_1	x_2	s_1	s_3	a_2	RHS	θ_i
0	s_1	0	$16/5$	1	$2/5$	0	8	$2.5 \rightarrow$
M	a_2	0	$6/5$	0	$1/5$	1	6	5
5	x_1	1	$2/5$	0	$-1/5$	0	2	5
	C_j	5	3	0	0	M		
	Z_j	5	$\frac{6M}{5} + 2$	0	$\frac{2M}{5} - 1$	M	$Z = 10 + 6M$	
	$C_j - Z_j$	0	$-\frac{6M}{5} + 1$	0	$-\frac{2M}{5} + 1$	0		

↑



Simplex Tableau: 3rd Iteration

c_B	Basis	x_1	x_2	s_1	s_3	a_2	RHS	θ_i
3	x_2	0	1	$5/16$	$1/8$	0	$5/2$	40
M	a_2	0	0	$-3/8$	$\left(\frac{1}{4}\right)$	1	3	$12 \rightarrow$
5	x_1	1	0	$-1/8$	$-1/4$	0	1	-
	$C_j \rightarrow$	5	3	0	0	M		
	Z_j	5	3	$\frac{5}{16} - \frac{3M}{8}$	$\frac{M}{4} - \frac{7}{8}$	M	$Z = \frac{25}{2} + 3M$	
	$C_j - Z_j$	0	0	$\frac{3M}{8} - \frac{5}{16}$	$-\frac{M}{4} + \frac{7}{8}$	0		

↑



Simplex Tableau: Final Iteration

c_B	Basis	x_1	x_2	s_1	s_3	RHS	θ_i
3	x_2	0	1	1/2	0	1	-
0	s_2	0	0	-3/2	1	12	-
5	x_1	1	0	-1/2	0	4	-
	$C_j \rightarrow$	5	3	0	0		
	Z_j	5	3	-1	0	$Z = 23$	
	$C_j - Z_j$	0	0	1	0		

As shown in the table above, all $c_j - z_j \geq 0$. Thus, the optimal solution is: $x_1 = 4, x_2 = 1, s_1 = 0, s_2 = 12$ and Minimum value is $Z = 23$.



Example Big-M Method

Maximize: $Z = x_1 + 2x_2 + 3x_3 - x_4$

Subject to:

$$x_1 + 2x_2 + 3x_3 = 15$$

$$2x_1 + x_2 + 5x_3 = 20$$

$$x_1 + 2x_2 + x_3 + x_4 = 10$$

Non-negativity: $x_1, x_2, x_3, x_4 \geq 0$



Solution

The standard form of the LP problem is:

Maximize: $Z = x_1 + 2x_2 + 3x_3 - x_4 - Ma_1 - Ma_2$

Subject to:

$$\begin{array}{rclcl} x_1 + 2x_2 + 3x_3 & +a_1 & = & 15 \\ 2x_1 + x_2 + 5x_3 & +a_2 & = & 20 \\ x_1 + 2x_2 + x_3 + x_4 & & = & 10 \end{array}$$

Non-negativity: $x_1, x_2, x_3, x_4, a_1, a_2 \geq 0$



Simplex Tableau

C_B	B	x_1	x_2	x_3	x_4	a_1	a_2	RHS	θ_i
-M	a_1	1	2	3	0	1	0	15	5
-M	a_2	2	1	⑤	0	0	1	20	4 $\rightarrow$
-1	x_4	1	2	1	1	0	0	10	10
	C_j	1	2	3	-1	-M	-M		
	Z_j	-3M-1	-3M-2	-8M-1	-1	-M	-M	$Z = -35M - 10$	
	$C_j - Z_j$	3M+2	3M+4	8M+4	0	0	0		

↑

Initial Basic Feasible Solution: $a_1 = 15, a_2 = 20, x_4 = 10$ and $x_1 = x_2 = x_3 = 0$.



Simplex Tableau II

C_B	B	x_1	x_2	x_3	x_4	a_1	RHS	θ_i
-M	a_1	-1/5	7/5	0	0	1	3	15/7 $\rightarrow$
3	x_3	2/5	1/5	1	0	0	4	20
-1	x_4	3/5	9/5	0	1	0	6	30/9
	C_j	1	2	3	-1	-M		
	Z_j	$\frac{M+3}{5}$	$\frac{-7M-6}{5}$	3	-1	-M	$Z = -3M + 6$	
	$C_j - Z_j$	$-\frac{M+2}{5}$	$\frac{7M+16}{5}$	0	0	0		

↑

Basic Feasible Solution: $a_1 = 3, x_3 = 4, x_4 = 6$ and $x_1 = x_2 = 0$.



Simplex Tableau III

C_B	B	x_1	x_2	x_3	x_4	RHS	Min Ratio
2	x_2	$-1/7$	1	0	0	$15/7$	—
3	x_3	$3/7$	0	1	0	$25/7$	$25/3$
-1	x_4	$6/7$	0	0	1	$15/7$	$15/6 \rightarrow$
	C_j	1	2	3	-1		
	Z_j	$1/7$	2	3	-1	$Z = 90/7$	
	$C_j - Z_j$	$6/7$	0	0	0		

↑



Optimal Solution

C_B	B	x_1	x_2	x_3	x_4	RHS
2	x_2	0	1	0	$1/6$	$15/6$
3	x_3	0	0	1	$-3/6$	$15/6$
1	x_1	1	0	0	$7/6$	$15/6$
	C_j	1	2	3	-1	
	Z_j	1	2	3	0	$Z = 15$
	$C_j - Z_j$	0	0	0	-1	

since all $c_j - z_j \leq 0$, therefore, an optimal solution is arrived at with value of variables as:

$$x_1 = \frac{15}{6}, x_2 = \frac{15}{6}, x_3 = \frac{15}{6} \text{ and Max } Z = 15.$$



Two-Phase Method

In the first phase of this method, the sum of the artificial variables is minimized subject to the given constraints in order to get a basic feasible solution of the LP problem.

The second phase minimizes the original objective function starting with the basic feasible solution obtained at the end of the first phase.

Since the solution of the LP problem is completed in two phases, this method is called the two-phase method.



Phase I

Step 1: (a) If all constraints are $\leq$ type, use Phase II directly. Otherwise, add surplus and artificial variables to convert to equality constraints.

(b) If the given LP problem is of minimization, then convert it to the maximization type by the usual method.

Step 2: Assign 0 coefficient to decision and surplus variables; assign -1 to artificial variables (a_i).

$$\text{Maximize } Z^* = \sum_{i=1}^m (-1)a_i$$

Subject to:

$$\sum_{j=1}^n c_{ij}x_j + ca_i = b_i, \quad i = 1, 2, \dots, m$$

$$x_j, a_i \geq 0$$



Phase I cont'd

Step 3: Apply the simplex algorithm to solve this auxiliary LP problem. The following three cases may arise at optimality:

(a) $\text{Max } Z^* < 0$ and at least one artificial variable is present in the basis with positive value. This means that no feasible solution exists for the original LP problem.



Phase I cont'd

Step 3: Apply the simplex algorithm to solve this auxiliary LP problem. The following three cases may arise at optimality:

(a) $\text{Max } Z^* < 0$ and at least one artificial variable is present in the basis with positive value. This means that no feasible solution exists for the original LP problem.

(b) $\text{Max } Z^* = 0$ and no artificial variable is present in the basis. This means that only decision variables (x_j 's) are present in the basis and hence proceed to Phase II to obtain an optimal basic feasible solution on the original LP problem.



Phase I cont'd

Step 3: Apply the simplex algorithm to solve this auxiliary LP problem. The following three cases may arise at optimality:

- (a) $\text{Max } Z^* < 0$ and at least one artificial variable is present in the basis with positive value. This means that no feasible solution exists for the original LP problem.
- (b) $\text{Max } Z^* = 0$ and no artificial variable is present in the basis. This means that only decision variables (x_j 's) are present in the basis and hence proceed to Phase II to obtain an optimal basic feasible solution on the original LP problem.
- (c) $\text{Max } Z^* = 0$ and at least one artificial variable is present in the basis at zero value. This means that a feasible solution to the auxiliary LP problem is also a feasible solution to the original LP problem. In order to arrive at the basic feasible solution, proceed directly to Phase II or else eliminate the artificial basic variable and then proceed to Phase II.



Phase II

Assign actual coefficients to the variables in the objective function and zero coefficient to the artificial variables which appear at zero value in the basis at the end of Phase I. The last simplex table of Phase I can be used as the initial simplex table for Phase II.

Then apply the usual simplex algorithm to the modified simplex table in order to get the optimal solution to the original problem. Artificial variables that do not appear in the basis may be removed.



Example

Use two-phase simplex method to solve the following LP problem:

Minimize $Z = x_1 + x_2$

subject to the constraints:

$$2x_1 + x_2 \geq 4$$

$$x_1 + 7x_2 \geq 7$$

and $x_1, x_2 \geq 0$



Solution

Converting the given LP problem objective function into the maximization form and then adding surplus variables s_1, s_2 and artificial variables a_1, a_2 in the constraints, the problem becomes:

Maximize $Z^* = -x_1 - x_2$

subject to the constraints:

$$\begin{array}{rclcl} 2x_1 + x_2 - s_1 + a_1 & & & = & 4 \\ x_1 + 7x_2 & & -s_2 + a_2 & = & 7 \end{array}$$

and $x_1, x_2, s_1, s_2, a_1, a_2 \geq 0$

where $Z^* = -Z$



Phase I

This phase starts by considering the following auxiliary LP problem:

Maximize $Z^* = -a_1 - a_2$

subject to the constraints:

$$2x_1 + x_2 - s_1 + a_1 = 4$$

$$x_1 + 7x_2 - s_2 + a_2 = 7$$

and $x_1, x_2, s_1, s_2, a_1, a_2 \geq 0$

The initial solution is presented in the Table in the next slide.



Phase 1

c_B	Basis	x_1	x_2	s_1	s_2	a_1	a_2	RHS	θ_i
-1	a_1	2	1	-1	0	1	0	4	4
-1	a_2	1	⑦	0	-1	0	1	7	$1 \rightarrow$
	c_j	0	0	0	0	-1	-1		
	z_j	-3	-8	1	1	-1	-1	-11	
	$c_j - z_j$	3	8	-1	-1	0	0		

↑



Phase I: Second Iteration

c_B	Basis	x_1	x_2	s_1	s_2	a_1	a_2	RHS	θ_i
-1	$a_1 \rightarrow$	13/7	0	-1	1/7	1	-1/7	3	$\frac{21}{13} \rightarrow$
0	x_2	1/7	1	0	-1/7	0	1/7	1	7
	c_j	0	0	0	0	-1	-1		
	z_j	-13/7	0	1	-1/7	-1	1/7	-3	
	$c_j - z_j$	13/7	0	-1	1/7	0	-8/7		

↑



Phase I: Final Iteration

c_B	Basis	x_1	x_2	s_1	s_2	RHS
0	x_1	1	0	$-7/13$	$1/13$	$21/13$
0	x_2	0	1	$1/13$	$-2/13$	$10/13$
	c_j	0	0	0	0	
	z_j	0	0	0	0	0
	$c_j - z_j$	0	0	0	0	

- ▶ Since all $c_j - z_j \leq 0$, the optimal solution:
 $x_1 = \frac{21}{13}, x_2 = \frac{10}{13}, s_1 = 0, s_2 = 21, a_1 = 0, a_2 = 0$ with $Z^* = 0$ is arrived at.
- ▶ Also $\text{Min } Z^* = 0$ and no artificial variable appears in the basis, hence this solution is the basic feasible solution to the original LP problem.
- ▶ Thus, we have to move to Phase II to get an optimal solution to our original LP problem.



Phase II: Initial Simplex Tableau

c_B	Basis	x_1	x_2	s_1	s_2	RHS
-1	x_1	1	0	$-7/13$	$-7/13$	$21/13$
-1	x_2	0	1	$1/13$	$1/13$	$10/13$
	c_j	-1	-1	0	0	
	z_j	-1	-1	$6/13$	$6/13$	$-31/13$
	$c_j - z_j$	0	0	$-6/13$	$-6/13$	

- ▶ Since all $c_j - z_j \leq 0$, the current solution is optimal.
- ▶ The optimal basic feasible solution is $x_1 = \frac{21}{13}, x_2 = \frac{10}{13}$.
- ▶ The resulting objective value is $\text{Max } Z^* = -\frac{31}{13}$, and $\text{Min } Z = -(\text{Max } Z^*) \therefore \text{Min } Z = \frac{31}{13}$.



Example 2: Two-Phase Simplex Method

Solve the following LP problem by using the two-phase simplex method:

Minimize

$$Z = x_1 - 2x_2 - 3x_3$$

Subject to the constraints:

$$-2x_1 + x_2 + 3x_3 = 2$$

$$2x_1 + 3x_2 + 4x_3 = 1$$

$$x_1, x_2, x_3 \geq 0$$



Solution

After converting the objective function into the maximization form and by adding artificial variables a_1 and a_2 in the constraints, the given LP problem becomes:

Maximize

$$Z^* = -x_1 + 2x_2 + 3x_3$$

Subject to the constraints:

$$-2x_1 + x_2 + 3x_3 + a_1 = 2$$

$$2x_1 + 3x_2 + 4x_3 + a_2 = 1$$

$$x_1, x_2, x_3, a_1, a_2 \geq 0$$

where $Z^* = -Z$.



Phase I: Auxiliary LP Problem

This phase starts by considering the following auxiliary LP problem:

Maximize

$$Z^* = -a_1 - a_2$$

Subject to the constraints:

$$-2x_1 + x_2 + 3x_3 + a_1 = 2$$

$$2x_1 + 3x_2 + 4x_3 + a_2 = 1$$

$$x_1, x_2, x_3, a_1, a_2 \geq 0$$

The initial solution is presented in the next table.



Phase I: Initial Simplex Tableau

c_B	Basis	x_1	x_2	x_3	a_1	a_2	b	θ_i
-1	a_1	-2	1	3	1	0	2	$2/3$
-1	$a_2 \rightarrow$	2	3	④	0	1	1	$1/4 \rightarrow$
	c_j	0	0	0	-1	-1		
	z_j	0	-4	-7	-1	-1	-3	
	$c_j - z_j$	0	4	7	0	0		

↑

- Initial Basic Solution: $x_1 = 0, x_2 = 0, x_3 = 0, a_1 = 2, a_2 = 1$ with $Z^* = -3$.



Phase I: Final Iteration

c_B	Basis	x_1	x_2	x_3	a_1	RHS
-1	a_1	-7/2	-5/4	0	1	5/4
0	x_3	1/2	3/4	1	0	1/4
	c_j	0	0	0	-1	
	z_j	7/2	5/4	0	-1	-5/4
	$c_j - z_j$	-7/2	-5/4	0	0	

- ▶ Since all $c_j - z_j \leq 0$, the optimal solution is reached.
- ▶ The optimal solution is $x_1 = 0, x_2 = 0, x_3 = \frac{1}{4}, a_1 = \frac{5}{4}, a_2 = 0$ with Max $Z^* = -\frac{5}{4}$.
- ▶ Because $\text{Max } Z^* < 0$ and the artificial variable a_1 appears in the basis with a positive value ($\frac{5}{4}$), a feasible solution to the given original LP problem does not exist.



DUALITY AND SENSITIVITY ANALYSIS (POST OPTIMALITY TEST)

DUALITY

Finding an optimal solution to a problem is not the only objective but the decision maker is interested in obtaining relevant information regarding the impact of technological innovations , economics of resource utilization and the sensitivity of the optimum solution to the changes in the model parameters. Such pertinent issues are handled by duality and sensitivity analysis or post optimality test (analysis).

Note: Every primal problem has its corresponding dual problem.



Characteristics of the Primal and Dual Problem

- (1) If the primal problem is a maximization problem, the dual is a minimization problem, and vice versa.
- (2) The profit constants (C_j) in the primal problem replace the capacity constants (resource b_j R.H.S) and vice versa.
- (3) If the primal problem involves $\leq$, the dual problem involves $\geq$, and vice versa.
- (4) In the constraint inequalities, the coefficients which were found by going from left to right are positioned in the dual from top to the bottom, and vice versa.
- (5) A new set of variables appear in the dual problem.
- (6) Neglecting the number of non-negativity conditions, if there are n variables and m inequalities in the primal problem, then the dual problem will have m variables and n inequalities.
- (7) The dual of the dual problem is the primal problem.



Primal Problem Definition

Let the primal problem be,

$$\text{Max } Z = C_1x_1 + C_2x_2 + \cdots + C_nx_n$$

Subject to,

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \leq b_m$$

$$x_1, x_2, \dots, x_n \geq 0$$



The Dual Problem

The dual problem is defined as,

$$\begin{aligned} \text{Min} \quad & Z' = b_1y_1 + b_2y_2 + \cdots + b_my_m \\ \text{Subject to,} \quad & \\ & a_{11}y_1 + a_{21}y_2 + \cdots + a_{m1}y_m \geq C_1 \\ & a_{12}y_1 + a_{22}y_2 + \cdots + a_{m2}y_m \geq C_2 \\ & \vdots \\ & a_{1n}y_1 + a_{2n}y_2 + \cdots + a_{mn}y_m \geq C_n \\ & y_1, y_2, \dots, y_m \geq 0 \end{aligned}$$

where, $y_1, y_2, \dots, y_m$ are called dual variables.



Duality Theorem

- ▶ If either the primal or the dual problem has a finite optimum then the other has a finite optimum solution.
- ▶ If either problem has an unbounded solution then the other problem has no feasible solution.
- ▶ The simplex procedure will give us the optimum solution if it exist for both primal and its dual problem simultaneously.



Example 1: Writing the Dual Problem

Primal:

$$\begin{aligned} \text{Maximize: } Z &= 45X_1 + 80X_2 \\ \text{s.t. } 5X_1 + 20X_2 &\leq 400 \\ 10X_1 + 15X_2 &\leq 450 \\ X_1, X_2 &\geq 0 \end{aligned}$$

Dual:



Example 1: Writing the Dual Problem

Primal:

$$\begin{aligned} \text{Maximize: } Z &= 45X_1 + 80X_2 \\ \text{s.t. } 5X_1 + 20X_2 &\leq 400 \\ 10X_1 + 15X_2 &\leq 450 \\ X_1, X_2 &\geq 0 \end{aligned}$$

Dual:

$$\begin{aligned} \text{Minimize: } Z &= 400y_1 + 450y_2 \\ \text{s.t. } 5y_1 + 10y_2 &\geq 45 \\ 20y_1 + 15y_2 &\geq 80 \\ y_1, y_2 &\geq 0 \end{aligned}$$

Where y_1 and y_2 are the dual variables.



Example 2: Writing the Dual Problem

Primal Problem:

$$\text{Minimize: } F = 28X_1 + 40X_2 + 20X_3$$

s.t

$$X_1 + 0.9X_2 + 0.8X_3 \geq 30$$

$$0.6X_1 + 2X_2 + 0.7X_3 \geq 23$$

$$X_1, X_2, X_3 \geq 0$$



Example 2: Writing the Dual Problem

Primal Problem:

$$\text{Minimize: } F = 28X_1 + 40X_2 + 20X_3$$

s.t

$$X_1 + 0.9X_2 + 0.8X_3 \geq 30$$

$$0.6X_1 + 2X_2 + 0.7X_3 \geq 23$$

$$X_1, X_2, X_3 \geq 0$$

Corresponding Dual Problem:

$$\text{Maximize: } Z = 30y_1 + 23y_2$$

s.t

$$y_1 + 0.6y_2 \leq 28$$

$$0.9y_1 + 2y_2 \leq 40$$

$$0.8y_1 + 0.7y_2 \leq 20$$

$$y_1, y_2 \geq 0$$



Example 3: Writing the Dual Problem

Write the dual of the following primal LP problem:

$$\begin{aligned} \text{Max} \quad & Z = x_1 + 2x_2 + x_3 \\ \text{Subject to,} \\ & 2x_1 + x_2 - x_3 \leq 2 \\ & -2x_1 + x_2 - 5x_3 \geq -6 \\ & 4x_1 + x_2 + x_3 \leq 6 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$



Solution

Since the problem is not in the canonical form, we interchange the inequality of the second constraint.

$$\text{Max } Z = x_1 + 2x_2 + x_3$$

Subject to,

$$2x_1 + x_2 - x_3 \leq 2$$

$$2x_1 - x_2 + 5x_3 \leq 6$$

$$4x_1 + x_2 + x_3 \leq 6$$

$$\text{and, } x_1, x_2, x_3 \geq 0$$



Solution cont'd

Dual: Let w_1, w_2, w_3 be the dual variables.

$$\text{Min } Z' = 2w_1 + 6w_2 + 6w_3$$

Subject to,

$$2w_1 + 2w_2 + 4w_3 \geq 1$$

$$w_1 - w_2 + w_3 \geq 2$$

$$-w_1 + 5w_2 + w_3 \geq 1$$

$$w_1, w_2, w_3 \geq 0$$



Example 4

Obtain the dual LP problem of the following primal LP problem:

$$\text{Minimize } Z = x_1 + 2x_2$$

subject to the constraints

$$2x_1 + 4x_2 \leq 160$$

$$x_1 - x_2 = 30$$

$$x_1 \geq 10$$

$$\text{and } x_1, x_2 \geq 0$$



Solution

Since the objective function of the primal LP problem is of minimization, change all $\leq$ type constraints to $\geq$ type constraints by multiplying the constraint on both sides by -1 . Also, write the $=$ type constraint equivalent to two constraints of the type $\geq$ and $\leq$. Then the given primal LP problem can be rewritten as:

Minimize: $Z = x_1 + 2x_2$

Subject to the constraints:

$$-2x_1 - 4x_2 \geq -160$$

$$x_1 - x_2 \geq 30$$

$$-x_1 + x_2 \geq -30$$

$$x_1 \geq 10$$

and $x_1, x_2 \geq 0$



Solution cont'd

Let y_1, y_2, y_3 , and y_4 be the dual variables corresponding to the four constraints in the given order. The dual of the given primal LP problem can then be formulated as follows:

$$\text{Maximize: } Z = -160y_1 + 30y_2 - 30y_3 + 10y_4$$

Subject to the constraints:

$$-2y_1 + y_2 - y_3 + y_4 \leq 1$$

$$-4y_1 - y_2 + y_3 \leq 2$$

$$\text{and } y_1, y_2, y_3, y_4 \geq 0$$



Solution cont'd

Let $y = y_2 - y_3$ where $y_2, y_3 \geq 0$. The previously formulated dual problem then reduces to the following form:

Maximize: $Z_y = -160y_1 + 30y + 10y_4$

Subject to the constraints:

$$-2y_1 + y + y_4 \leq 1$$

$$-4y_1 - y \leq 2$$

and $y_1, y_4 \geq 0$; y is unrestricted in sign



Unrestricted Variables

- ▶ In actual practice, decision variables, x_j , $j = 1, 2, \dots, n$ should have non-negative values.
- ▶ However, in many situations, one or more of these variables may have either positive, negative or zero value.
- ▶ Variables that can assume positive, negative or zero value are called unrestricted variables.



Unrestricted Variable

- ▶ The use of the simplex method requires that all the decision variables must have non-negative value at each iteration.
- ▶ Therefore in order to convert an LP problem involving unrestricted variables into an equivalent LP problem having only non-negative variables, each of unrestricted variable is expressed as the difference of two non-negative variables.



Unrestricted Variable Cont...

- ▶ Let variable x_r be unrestricted in sign. We define two new variables say x'_r and x''_r such that

$$x_r = x'_r - x''_r; \quad x'_r, x''_r \geq 0$$

- ▶ If $x'_r \geq x''_r$, then $x_r \geq 0$, but if $x'_r \leq x''_r$, then $x_r \leq 0$. Also if $x'_r = x''_r$, then $x_r = 0$.
- ▶ Hence depending on the value of x'_r and x''_r , the variable x_r can have either positive or negative sign.



Unrestricted Variables Contd...

For example, the following LP problem:

$$\text{Maximize } Z = \sum_{j \neq r}^n c_j x_j + c_r x_r$$

subject to the constraints:

$$\sum_{j \neq r}^n a_{ij} x_j + a_{ir} x_r = b_i, \quad i = 1, 2, \dots, m$$

and $x_j \geq 0$, and x_r are unrestricted in sign; $j = 1, 2, \dots, n, \quad j \neq r$.



Conversion to Standard Form

These variables can be converted into its equivalent standard form as follows:

$$\text{Maximize } Z = \sum_{j \neq r}^n c_j x_j + c_r(x'_r - x''_r)$$

subject to the constraints:

$$\sum_{j \neq r}^n a_{ij} x_j + a_{ir}(x'_r - x''_r) = b_i, \quad i = 1, 2, \dots, m$$

$$\text{and } x_j, x'_r, x''_r \geq 0, \quad j = 1, 2, \dots, n, \quad j \neq r$$



Optimal Solution Analysis

- ▶ Variables x_r' and x_r'' simultaneously cannot appear in the basis (since the column vectors corresponding to these variables are linearly dependent).
- ▶ Thus any of the following three cases may arise at the optimal solution:
 1. $x_r' = 0 \rightarrow x_r = -x_r''$
 2. $x_r'' = 0 \rightarrow x_r = x_r'$
 3. $x_r' = x_r'' = 0 \rightarrow x_r = 0$
- ▶ This indicates that the value of x_r' and x_r'' uniquely determines the values of the variable x_r .



Example

Use the simplex method to solve the following LP problem.

$$\text{Maximize } Z = 3x_1 + 2x_2 + x_3$$

subject to the constraints:

$$2x_1 + 5x_2 + x_3 = 12$$

$$3x_1 + 4x_2 = 11$$

$$\text{and } x_2, x_3 \geq 0 \text{ and}$$

x_1 unrestricted in sign



Standard Form Conversion

Introducing an artificial variable a_2 in the second constraint of the given LP problem in order to obtain the basis matrix. Since x_1 is unrestricted in sign, introducing the non-negative variables x'_1 and x''_1 so that $x_1 = x'_1 - x''_1$ where $x'_1, x''_1 \geq 0$. The standard form of the LP problem now becomes:

$$\text{Maximize } Z = 3(x'_1 - x''_1) + 2x_2 + x_3 - Ma_2$$

subject to the constraints:

$$2(x'_1 - x''_1) + 5x_2 + x_3 = 12$$

$$3(x'_1 - x''_1) + 4x_2 + a_2 = 11$$

$$\text{and } x'_1, x''_1, x_2, x_3, a_2 \geq 0$$



Initial Solution

The initial solution is shown in the Table below

C_B	Basis	x'_1	x''_1	x_2	x_3	a_2	RHS	Min Ratio
1	x_3	2	-2	5	1	0	12	
-M	a_2	3	-3	4	0	1	11	
	c_j	3	-3	2	1	-M		
	z_j							
	$c_j - z_j$							



Simplex tableau

C_B	Basis	x'_1	x''_1	x_2	x_3	a_2	RHS	Min Ratio
1	x_3	2	-2	⑤	1	0	12	$2.4 \rightarrow$
-M	a_2	3	-3	4	0	1	11	2.75
	c_j	3	-3	2	1	-M		
	z_j	$-3M+2$	$3M-2$	$-4M+5$	1	-M	$Z = -11M + 12$	
	$c_j - z_j$	$3M+1$	$-3M-1$	$4M-3$	0	0		

↑



Next Iteration

Pivoting x_2 into the basis:

C_B	Basis	x'_1	x''_1	x_2	x_3	a_2	RHS	Min Ratio
2	x_2	$2/5$	$-2/5$	1	$1/5$	0	$12/5$	$(12/5)/(2/5) = 6$
-M	a_2	$(7/5)$	$-7/5$	0	$-4/5$	1	$7/5$	$(7/5)/(7/5) = 1 \rightarrow$
	c_j	3	-3	2	1	-M		
	z_j	$\frac{4-7M}{5}$	$\frac{-4+7M}{5}$	2	$\frac{2-4M}{5}$	-M	$Z = \frac{24-7M}{5}$	
	$c_j - z_j$	$\frac{11+7M}{5}$	$\frac{-11-7M}{5}$	0	$\frac{3+4M}{5}$	0		

↑



Next Iteration

C_B	Basis	x'_1	x''_1	x_2	x_3	a_2	RHS	Min Ratio
2	x_2	0	0	1	$\frac{3}{7}$	$-\frac{2}{7}$	2	$14/3 \rightarrow$
3	x'_1	1	-1	0	$-\frac{4}{7}$	$\frac{5}{7}$	1	-
	c_j	3	-3	2	1	-M		
	z_j	3	-3	2	$-\frac{6}{7}$	$\frac{11}{7}$	$Z = 7$	
	$c_j - z_j$	0	0	0	$\frac{13}{7}$	$-M - \frac{11}{7}$		

↑



Next Iteration

Pivoting x_3 into the basis replacing x_2 :

C_B	Basis	x'_1	x''_1	x_2	x_3	a_2	RHS	Min Ratio
1	x_3	0	0	$7/3$	1	$-2/3$	$14/3$	-
3	x'_1	1	-1	$4/3$	0	$3/7$	$11/3$	-
	c_j	3	-3	2	1	-M		
	z_j	3	-3	$19/3$	1	$5/7$	$Z = 25/3$	
	$c_j - z_j$	0	0	$-13/3$	0	$-M - 5/7$		

Since in the table above all $c_j - z_j \leq 0$, we get an optimal solution:

$x'_1 = \frac{11}{3}$ and $x_1 = x'_1 - x''_1 = \frac{11}{3} - 0 = \frac{11}{3}$; $x_2 = 0$ and $x_3 = \frac{14}{3}$ with $\text{Max } Z = \frac{25}{3}$.



Sensitivity Analysis

After a linear programming problem has been solved, it is useful to study the effects of instantaneous changes in the parameters of the problem on the current optimal solution.

For example, after solving the linear programming problem, one may find that certain values of the variables have changed and therefore will like to know whether the current optimal solution is still optimal under the changed circumstances. Or, one may want to know how far the input parameter values can vary without causing any changes in the optimal solution or the composition of a basis set.

Such an investigation is known as Sensitivity Analysis or Post-Optimality Test (Analysis) in linear programming.



Sensitivity Analysis cont'd...

What happens when the resource availability is curtailed or increased?

What happens if a new activity is added or if an existing activity is deleted?

NOTE:

The strength of the simplex method is that the final tableau gives answers to some of these specific post-optimality questions.



Example

Maximize:

$$Z = 3x_1 + 3x_2 + 4x_3 + 7x_4$$

Subject to (s.t):

$$x_1 + 3x_2 + x_3 + x_4 \leq 9$$

$$5x_1 + 3x_2 + 2x_3 + x_4 \leq 60$$

$$x_1 + 3x_2 + 5x_3 + 8x_4 \leq 50$$

$$x_1, x_2, x_3, x_4 \geq 0$$



Final Solution

The final solution is shown in the table below:

C_B	Basis	x_1	x_2	x_3	x_4	s_1	s_2	s_3	R.H.S
3	x_1	1	$5/7$	$3/7$	0	$8/7$	0	$-1/7$	$22/7$
0	s_2	0	$-6/7$	$-5/7$	0	$-39/7$	1	$4/7$	$269/7$
7	x_4	0	$2/7$	$4/7$	1	$-1/7$	0	$4/28$	$41/7$
	C_j	3	3	4	7	0	0	0	
	Z_j	3	$29/7$	$37/7$	7	$17/7$	0	$4/7$	$353/7$
	$C_j - Z_j$	0	$-8/7$	$-9/7$	0	$-17/7$	0	$-4/7$	



Optimal Solution and Sensitivity

Optimal Solution

$$x_1 = \frac{22}{7}, \quad x_2 = 0, \quad x_3 = 0, \quad x_4 = \frac{41}{7}$$

$$\text{and MaxZ} = \frac{353}{7}$$

Changes in the Coefficient of Objective Function - Range of Optimality

These changes are studied by grouping them into two categories:

1. C_j 's corresponding to non-basic variables.
2. C_j 's corresponding to basic variables.



Non-basic Variables Analysis

The non-basic variables in the above problem are x_2, x_3, s_1, s_3 . By looking at the final tableau, we find that if C_2 is increased beyond $\frac{8}{7}$, then $C_2 - Z_2$ will become positive and hence x_2 will enter the basis.

Thus, the coefficients of the $C_j - Z_j$ row in the final simplex tableau for non-basic variables indicate the extent of increase in their respective coefficients of the objective function while still keeping the current solution optimal.

If these coefficients are increased beyond this limit, the current optimal solution will change.



Basic Variables Analysis

In the above final tableau, x_1 and x_4 are basic variables and we are interested in knowing the extent of changes in their coefficients in the objective function.

Consider the basic variable x_1 and define Δ_1 as the change in C_1 . In order to find out the range for Δ_1 , we each divide the $C_j - Z_j$ coefficient corresponding to non-basic variables by the corresponding coefficient in the x_1 row.



Sensitivity Analysis cont'd...

This is given as in the table below:

x_1	1	$5/7$	$3/7$	0	$8/7$	0	$-1/7$
$C_j - Z_j$	0	$-8/7$	$-9/7$	0	$-17/7$	0	$-4/7$
$(C_j - Z_j)/x_1$	-	$-8/5$	$-9/3$	-	$-17/8$	-	4



Sensitivity Analysis cont'd...

The positive coefficient in the $(C_j - Z_j)/x_1$ row gives the upper limit for Δ_1 .
NOTE:

- ▶ If there are more than one positive coefficient, select the one with the smallest coefficient.
- ▶ For the lower limit, select among the negative coefficients the one which has the lowest absolute value.

$$\frac{-8}{5} \leq \Delta_1 \leq 4$$

After that, go to the original problem and find the coefficient of x_1 :

$$3 - \frac{8}{5} \leq C_1 \leq 3 + 4$$

$$\frac{7}{5} \leq C_1 \leq 7$$



Range of Feasibility - Changes in Right Hand Side

To determine the range of feasibility, we divide every coefficient in the right-hand side (R.H.S) column by the corresponding coefficient in the column of the slack variable associated with that specific resource constraint.

These computations are illustrated as follows:



Sensitivity Analysis cont'd...

R.H.S	S_1	R.H.S/ S_1
22/7	8/7	22/8
269/7	-39/7	-269/39
41/7	-1/7	-41

The least positive ratio indicates the extent by which this resource can be reduced without affecting the current solution. Also, the least negative ratio in absolute value ($-\frac{269}{39}$) indicates the extent to which this resource can be increased without affecting the current solution.

$$\left(9 - \frac{22}{8}, 9 + \frac{269}{39}\right)$$
$$\left(\frac{25}{4}, \frac{620}{39}\right)$$



Types of Linear Programming Solutions

While solving any LP problem using the simplex method, at the stage of the optimal solution, the following three types of solutions may exist:

1. Alternative (Multiple) Optimal Solution
2. Unbounded Solutions
3. Infeasible Solutions



Alternative (Multiple) Optimal Solution

The $c_j - z_j$ values in the simplex table indicate the contribution to the objective function value by each unit of a variable chosen to enter the basis.

An optimal solution to a maximization LP problem is reached when $c_j - z_j \leq 0$.

However, if $c_j - z_j = 0$ for a non-basic variable column in the optimal simplex table, and that variable is chosen to enter the basis, then another optimal solution is obtained which shows no improvement in the value of the objective function.



Example : (Alternative (Multiple) Optimal Solution)

Solve the following LP problem Maximize:

$$Z = 6x_1 + 4x_2$$

Subject to the constraints:

$$2x_1 + 3x_2 \leq 30$$

$$3x_1 + 2x_2 \leq 24$$

$$x_1 + x_2 \geq 3$$

$$\text{and } x_1, x_2 \geq 0$$



Solution

Maximize:

$$Z = 6x_1 + 4x_2 + 0s_1 + 0s_2 + 0s_3 - Ma_3$$

Subject to the constraints:

$$2x_1 + 3x_2 + s_1 = 30$$

$$3x_1 + 2x_2 + s_2 = 24$$

$$x_1 + x_2 - s_3 + a_3 = 3$$

$$\text{and } x_1, x_2, s_1, s_2, s_3, a_3 \geq 0$$



Simplex Tableau

C_B	Basis	x_1	x_2	s_1	s_2	s_3	a_3	RHS
0	s_1	2	3	1	0	0	0	30
0	s_2	3	2	0	1	0	0	24
-M	a_3	1	1	0	0	-1	1	3
	C_j	6	4	0	0	0	-M	
	Z_j							
	$C_j - Z_j$							



Initial Simplex Tableau

The initial solution is shown in the table below:

C_B	Basis	x_1	x_2	s_1	s_2	s_3	a_3	RHS	Min Ratio
0	s_1	2	3	1	0	0	0	30	$30/2 = 15$
0	s_2	3	2	0	1	0	0	24	$24/3 = 8$
-M	$a_3 \rightarrow$	①	1	0	0	-1	1	3	$3/1 = 3 \rightarrow$
	C_j	6	4	0	0	0	-M		
	Z_j	-M	-M	0	0	M	-M		
	$C_j - Z_j$	$6+M$	$4+M$	0	0	-M	0		

↑



Simplex Tableau: Iteration 2

The second iteration table is shown below:

C_B	Basis	x_1	x_2	s_1	s_2	s_3	a_3	RHS	Min Ratio
0	s_1	0	1	1	0	2	-2	24	$24/2 = 12$
0	$s_2 \rightarrow$	0	-1	0	1	3	-3	15	$15/3 = 5 \rightarrow$
6	x_1	1	1	0	0	-1	1	3	-
	C_j	6	4	0	0	0	-M		
	Z_j	6	6	0	0	-6	6	18	
	$C_j - Z_j$	0	-2	0	0	6	-M-6		

↑



Simplex Tableau: Iteration 3 (Optimal solution)

The final iteration table is shown below:

C_B	Basis	x_1	x_2	s_1	s_2	s_3	a_3	RHS
0	s_1	0	$5/3$	1	$-2/3$	0	0	14
0	s_3	0	$-1/3$	0	$1/3$	1	-1	5
6	x_1	1	$2/3$	0	$1/3$	0	0	8
	C_j	6	4	0	0	0	-M	
	Z_j	6	4	0	2	0	0	48
	$C_j - Z_j$	0	0	0	-2	0	-M	

Since all $C_j - Z_j \leq 0$ an optimal solution is reached and the optimal solution is $x_1 = 8$, $x_2 = 0$, $s_1 = 14$, $s_2 = 0$, $s_3 = 5$ with an optimal value $Z_{\max} = 48$



Simplex Tableau: Alternative Solution

Since $C_j - Z_j = 0$ corresponds to a non-basic variable x_2 . Thus, an alternative optimal solution can also be obtained by entering variable x_2 into the basis and removing basic variable, s_1 from the basis. The new solution is shown in below

C_B	Basis	x_1	x_2	s_1	s_2	s_3	a_3	RHS	Min Ratio
0	$s_1 \rightarrow$	0	5/3	1	-2/3	0	0	14	$14/(5/3) = 8.4$
0	s_3	0	-1/3	0	1/3	1	-1	5	-
6	x_1	1	2/3	0	1/3	0	0	8	$8/(2/3) = 12$
	C_j	6	4	0	0	0	-M		
	Z_j	6	4	0	2	0	0	48	
	$C_j - Z_j$	0	0	0	-2	0	-M		
			↑						



Simplex Tableau: Alternative Optimal Solution

After pivoting x_2 into the basis, the alternative optimal solution is:

C_B	Basis	x_1	x_2	s_1	s_2	s_3	a_3	RHS
4	x_2	0	1	$3/5$	$-2/5$	0	0	$42/5$
0	s_3	0	0	$1/5$	$1/5$	1	-1	$39/5$
6	x_1	1	0	$-2/5$	$3/5$	0	0	$12/5$
	C_j	6	4	0	0	0	-M	
	Z_j	6	4	0	2	0	0	48
	$C_j - Z_j$	0	0	0	-2	0	-M	

The alternative optimal solution is $x_1 = \frac{12}{5}$, $x_2 = \frac{42}{5}$, $s_1 = 0, s_2 = 0$, $s_3 = \frac{39}{5}$ with $Z_{\max} = 48$.



Unbounded Solution

In a maximization LP problem, if $c_j - z_j > 0$, ($c_j - z_j < 0$ for a minimization case) corresponds to a non-basic variable column in simplex table, and all a_{ij} values in this column are negative value in denominator would indicate the entry of a non-basic variable in the basis with a negative value (an infeasible solution).

Also, a zero value in the denominator would result in a ratio having an infinite value and would indicate that the value of non-basic variable would be increased infinitely with any of the current basic variables being removed from the basis.



Example : Unbounded Solution

Solve the following LP problem:

Maximize: $Z = 3x_1 + 5x_2$

Subject to:

$$x_1 - 2x_2 \leq 6$$

$$x_1 \leq 10$$

$$x_2 \geq 1$$

$$x_1, x_2 \geq 0$$



Converting to Standard Form

Minimize

$$Z = 3x_1 + 5x_2 + 0s_1 + 0s_2 + 0s_3 - M_{23}$$

Subject to

$$x_1 - 2x_2 + s_1 = 6$$

$$x_1 + s_2 = 10$$

$$x_2 - s_3 + a_3 = 1$$

Non-negativity Constraints:

$$x_1, x_2, s_1, s_2, s_3, a_3 \geq 0$$



Simplex Tableau

C_B	Basis	x_1	x_2	s_1	s_2	s_3	a_3	RHS	Min Ratio
0	s_1	1	-2	1	0	0	0	6	
0	s_2	1	0	0	1	0	0	10	
-M	a_3	0	1	0	0	-1	1	1	
	C_j	3	5	0	0	0	-M		
	Z_j								
	$C_j - Z_j$								



Initial Simplex Tableau

Simplex Tableau

C_B	Basis	x_1	x_2	s_1	s_2	s_3	a_3	RHS	Min Ratio
0	s_1	1	-2	1	0	0	0	6	-
0	s_2	1	0	0	1	0	0	10	-
-M	$a_3 \rightarrow$	0	1	0	0	-1	1	1	1 $\rightarrow$
	C_j	3	5	0	0	0	-M		
	Z_j	0	-M	0	0	M	-M	-M	
	$C_j - Z_j$	3	5+M	0	0	-M	0		

↑



Simplex Tableau (Iteration 2)

C_B	Basis	x_1	x_2	s_1	s_2	s_3	a_3	RHS
0	s_1	1	0	1	0	-2	2	8
0	s_2	1	0	0	1	0	0	10
5	x_2	0	1	0	0	-1	1	1
	Z_j	0	5	0	0	-5	5	5
	$C_j - Z_j$	3	0	0	0	5	-M-5	



Unbounded Solution

In the table above, $c_2 - z_2 = 5$ is the largest positive, so variable s_3 should enter into the basis. But, coefficients in the s_3 column are all negative or zero.

This indicates that s_3 cannot be entered into the basis. However, the value of s_3 can be increased infinitely without removing any one of the basic variables. Further, since s_3 is associated with x_2 in the third constraint, x_2 will also be increased infinitely because it can be expressed as $x_2 = 1 + s_3 - a_3$. Hence, the solution to the given problem is unbounded.



Infeasible Solution

If LP problem solution does not satisfy all of the constraints, then such a solution is called infeasible solution.

Also, infeasible solution occurs when all $c_j - z_j$ values satisfy optimal solution condition but at least one of artificial variables appears in the basis with a positive value. This situation may occur when an LP model is either improperly formulated or more than two of the constraints are incompatible.



Example

Solve the following LP problem

Maximize $Z = 6x_1 + 4x_2$

subject to the constraints

$$x_1 + x_2 \leq 5$$

$$x_2 \geq 8$$

$$\text{and } x_1, x_2 \geq 0$$



Standard Form of LP Problem

Standard Form

Adding slack, surplus and artificial variables, the standard form of LP problem becomes

$$\text{Maximize } Z = 6x_1 + 4x_2 + 0s_1 + 0s_2 - Ma_1$$

subject to the constraints

$$x_1 + x_2 + s_1 = 5$$

$$x_2 - s_2 + a_2 = 8$$

$$\text{and } x_1, x_2, s_1, s_2, a_2 \geq 0$$



Simplex Tableau

C_B	Basis	x_1	x_2	s_1	s_2	a_2	RHS	Min Ratio
0	s_1	1	1	1	0	0	5	
-M	a_2	0	1	0	-1	1	8	
	C_j	6	4	0	0	-M		
	Z_j							
	$C_j - Z_j$							



Initial Simplex Tableau

C_B	Basis	x_1	x_2	s_1	s_2	a_2	RHS	Min Ratio
0	$s_1 \rightarrow$	1	1	1	0	0	5	$5/1 = 5 \rightarrow$
-M	a_2	0	1	0	-1	1	8	$8/1 = 8$
	C_j	6	4	0	0	-M		
	Z_j	0	-M	0	M	-M	-8M	
	$C_j - Z_j$	6	$4+M$	0	-M	0		

↑



Iteration 2

C_B	Basis	x_1	x_2	s_1	s_2	a_2	RHS
4	x_2	1	1	1	0	0	5
-M	a_2	-1	0	-1	-1	1	3
	C_j	6	4	0	0	-M	
	Z_j	4+M	4	4+M	M	-M	20-3M
	$C_j - Z_j$	2-M	0	-4-M	-M	0	

Since all $c_j - z_j \leq 0$, the current solution is optimal. But this solution is not feasible for the given LP problem because the values of decision variables are: $x_1 = 0$ and $x_2 = 5$ violates second constraint, $x_2 \geq 8$.

The presence of artificial variable $a_2 = 3$ in the solution also indicates that the optimal solution violates the second constraint ($x_2 \geq 8$) by 3 units.



APPLICATION OF LINEAR PROGRAMMING

Transportation, Transshipment and Assignment models

1. In this lesson, we shall examine three special applications of the linear programming model, i.e: transportation, transshipment, and assignment models.
2. They are part of a larger class of linear programming problems known as network flow problems.
3. These problems have special mathematical characteristics that have enabled management scientists to develop very efficient, unique mathematical solution approaches to them.
4. These solution approaches are variations of the traditional simplex solution procedure.



THE TRANSPORTATION MODEL

Introduction

1. The transportation model is formulated for a class of problems with the following unique characteristics:
 - ▶ A product is transported from a number of sources to a number of destinations at the minimum possible cost; and
 - ▶ each source is able to supply a fixed number of units of the product, and each destination has a fixed demand for the product.
2. Although the general transportation model can be applied to a wide variety of problems, it is this particular application to the transportation of goods that is most familiar and from which the problem draws its name.



The Transportation Model

1. The following example demonstrates the formulation of the transportation model.

Example

Wheat is harvested in the Midwest and stored in grain elevators in three different cities namely; Kansas City, Omaha, and Des Moines. These grain elevators supply three flour mills, located in Chicago, St. Louis, and Cincinnati. Grain is shipped to the mills in railroad cars, each car capable of holding 1 ton of wheat.



Example Cont'd

- ▶ Table 1 shows the no. of tons (i.e., railroad cars) supplied from each elevator to the mills/month:

Table: Supply Constraints

Elevator	Supply
1. Kansas City	150
2. Omaha	175
3. Des Moines	275
Total	600 tons

- ▶ Table 2 shows demands from each mill (i.e., number of tons/month:

Table: Demand Data

Elevator	Demand
A. Chicago	200
B. St. Louis	100
C. Cincinnati	300
Total	600 tons

- ▶ Table 3 shows the cost to transport 1 ton from each elevator to each mill. This differs, according to the distance and rail system.

Table: Costs/unit/route

		Mills	
Elevator	A	B	C
1	\$6	\$8	\$10
2	7	11	11
3	4	5	12



The Transportation Model

Example Cont'd

- ▶ The problem is to determine how many tons to transport from each elevator to each mill/month to min. the total transportation cost.
- ▶ Figure 1 shows the different transportation routes, with supply and demand.

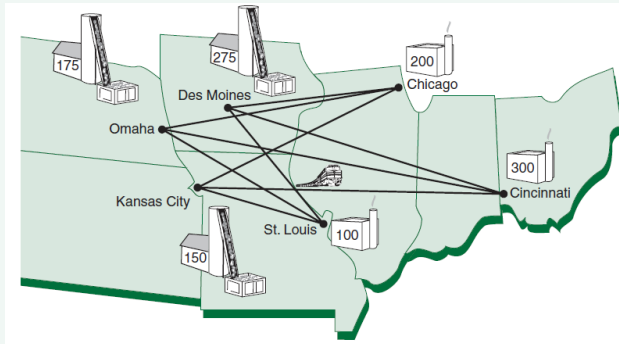


Figure: The Transportation Network.



The Transportation Model

Example Cont'd

- The linear programming model for this problem is formulated as follows:

$$\begin{aligned} \text{Minimize} \quad Z = & 6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + \\ & 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C} \end{aligned}$$

$$\begin{aligned} \text{Subject to} \quad & x_{1A} + x_{1B} + x_{1C} = 150 \\ \text{Supply const.:} \quad & x_{2A} + x_{2B} + x_{2C} = 175 \\ & x_{3A} + x_{3B} + x_{3C} = 275 \end{aligned}$$

$$\begin{aligned} \text{Demand const.:} \quad & x_{1A} + x_{2A} + x_{3A} = 200 \\ & x_{1B} + x_{2B} + x_{3B} = 100 \\ & x_{1C} + x_{2C} + x_{3C} = 300 \end{aligned}$$

$$\begin{aligned} \text{Non-negativity:} \quad & x_{ij} \geq 0 \end{aligned}$$



The Transportation Model

Example Cont'd

- ▶ In the above model, the decision variables, x_{ij} represent the number of tons of wheat transported from each grain elevator, i (where $i = 1, 2, 3$), to each mill, j (where $j = A, B, C$).
- ▶ The objective function represents the total transportation cost for each route.
- ▶ Each term in the objective function reflects the cost of the tonnage transported for one route.
- ▶ For example, if 20 tons are transported from elevator 1 to mill A, the cost (\$6) is multiplied by x_{1A} ($= 20$) which equals \$120.



The Transportation Model

Example Cont'd

- ▶ The first three constraints in the linear programming model represent the supply at each elevator; the last three constraints represent the demand at each mill.
- ▶ As an example, consider the first supply constraint $x_{1A} + x_{1B} + x_{1C} = 150$.
- ▶ This constraint represents the tons of wheat transported from Kansas City to all three mills: Chicago (x_{1A}) St. Louis (x_{1B}) and Cincinnati (x_{1C}).
- ▶ The amount transported from Kansas City is limited to the 150 tons available.



The Transportation Model

Example Cont'd

- ▶ Note that this constraint (as well as all others) is an equation ($=$) rather than a ($\leq$) inequality because all the tons of wheat available will be needed to meet the total demand of 600 tons.
- ▶ In other words, the three mills demand 600 total tons, which is the exact amount that can be supplied by the three grain elevators.
- ▶ Thus, all that can be supplied will be, in order to meet demand.
- ▶ This type of model, in which supply exactly equals demand, is referred to as a balanced transportation model.



The Transportation Model

Example Cont'd

- ▶ Realistically, however, an unbalanced problem, in which supply exceeds demand or demand exceeds supply, is a more likely occurrence.
- ▶ In our wheat transportation example, if the demand at Cincinnati is increased from 300 tons to 350 tons, a situation is created in which total demand is 650 tons and total supply is 600 tons



The Transportation Model

Example Cont'd

- ▶ One of the demand constraints will not be met because there is not enough total supply to meet total demand.
- ▶ If, instead, supply exceeds demand, then the supply constraints will be ($\leq$).
- ▶ Sometimes one or more of the routes in the transportation model may be prohibited.
- ▶ That is, units cannot be transported from a particular source to a particular destination.
- ▶ When this situation occurs, we must make sure that the variable representing that route does not have a value in the optimal solution.



The Transportation Model

Example Cont'd

- ▶ This can be accomplished by assigning a very large relative cost as the coefficient of this prohibited variable in the objective function.
- ▶ For example, in our wheat-shipping example, if the route from Kansas City to Chicago is prohibited (perhaps because of a rail strike), the variable x_{1A} is given a coefficient of 100 instead of 6 in the objective function, so x_{1A} will equal zero in the optimal solution because of its high relative cost.



The Transportation Model

Summary

- ▶ In a transportation problem, items are allocated from sources to destinations at a minimum cost.
- ▶ The linear programming model for a transportation problem has constraints for supply at each source and demand at each destination.
- ▶ In a balanced transportation model in which supply equals demand, all constraints are equalities.
- ▶ In an unbalanced transportation model, supply is greater than demand or demand is greater than supply



Transportation Solution Methods

- ▶ We shall use the balanced model in example 1 to showcase the solution of a transportation model.
- ▶ Transportation models are solved manually within a tableau, as in the simplex method.
- ▶ Each cell in a transportation tableau is analogous to a decision variable that indicates the amount allocated from a source to a destination.
- ▶ The tableau for the model in example 1 is shown below.



THE INITIAL TRANSPORTATION SIMPLEX TABLEAU

1	To			
From	A	B	C	Supply
1	6	8	10	150
2	7	11	11	
3	4	5	12	275
Demand	200	100	300	



- ▶ The supply and demand values along the outside rim of a tableau are called rim requirements.
- ▶ Transportation models are solved manually within a tableau, as in the simplex method.
- ▶ Transportation models do not start at the origin where all decision variables equal zero; they must be given an initial feasible solution from the initial simplex tableau.



Methods for finding Initial Basic Feasible Solutions

There are several methods for finding the initial feasible solution to transportation problems. However, we shall employ:

- ▶ North-west corner method
- ▶ The Minimum Cell Cost Method / Least Cost Method (LCM)
- ▶ Vogel's Mpproximation Method (VAM)
- ▶ Acheson Demand-Based Allocation Method (ADBAM)



The North-West Corner Method

- ▶ To find an initial solution by the north-west corner method,
 1. Allocate as much as possible to the cell in the upper left-hand corner, subject to the supply and demand constraints.
 2. Allocate as much as possible to the next adjacent feasible cell.
 3. Repeat step 2 until all rim requirements have been met.
- ▶ An initial solution is complete when all rim requirements are satisfied.
- ▶ The North-West Corner solution for the model example 1 is shown on the next slide with the total minimum cost in the bottom right corner of the tableau 2.



THE INITIAL TRANSPORTATION SIMPLEX TABLEAU

1	To			
From	A	B	C	Supply
1	6	8	10	150
2	7	11	11	
3	4	5	12	275
Demand	200	100	300	



THE NORTH-WEST CORNER SOLUTION

2A	To			
From	A	B	C	Supply
1	6	8	10	150 → 0
	150			
2	7	11	11	175
3	4	5	12	275
Demand	200 → 50	100	300	



THE NORTH-WEST CORNER SOLUTION

2B	To			
From	A	B	C	Supply
1	6 150	8	10	0
2	7 50	11	11	175 → 125
3	4	5	12	275
Demand	50 → 0	100	300	



THE NORTH-WEST CORNER SOLUTION

2C	To			
From	A	B	C	Supply
1	6 150	8	10	0
2	7 50	11 100	11	125 → 25
3	4	5	12	275
Demand	0	100 → 0	300	



THE NORTH-WEST CORNER SOLUTION

2D	To						
From	A		B		C		Supply
1	150	6		8		10	0
2	50	7	100	11	25	11	25 → 0
3		4		5		12	275
Demand	0		0		300 → 275		



THE NORTH-WEST CORNER SOLUTION

2E	To						
From	A		B		C		Supply
1	150	6		8		10	150
2	50	7	100	11	25	11	0
3		4		5	275	12	275 → 0
Demand	0		0		275 → 0		



THE NORTH-WEST CORNER SOLUTION

2F	To						
From	A		B		C		Supply
1	150	6		8		10	150
2	50	7	100	11	25	11	175
3		4		5	275	12	275
Demand	200		100		300		\$5,925



North-West Corner SOLUTION

ALLOCATED CELLS

The non-zero allocations from the North-West corner method are:

- ▶ $x_{1A} = 150$
- ▶ $x_{2A} = 50, \quad x_{2B} = 100, \quad x_{2C} = 25$
- ▶ $x_{3C} = 275$

TOTAL COST

The total cost Z is calculated as the sum of (unit cost $\times$ allocation):

$$Z = 6(x_{1A}) + 7(x_{2A}) + 11(x_{2B}) + 11(x_{2C}) + 12(x_{3C})$$

$$Z = 6(150) + 7(50) + 11(100) + 11(25) + 12(275) = \$5925$$



THE MINIMUM CELL COST METHOD

1. To find an initial solution by the Least Cost method,
 - 1.1 Allocate as much as possible to the feasible cell with the minimum transportation cost, and adjust the rim requirements.
 - 1.2 Repeat step 1 until all rim requirements have been met.
2. An initial solution is complete when all rim requirements are satisfied.
3. The Least Cost solution for the above model example 1 is shown on the next slide with the total minimum cost in the bottom right corner of the tableau 3.



THE INITIAL TRANSPORTATION SIMPLEX TABLEAU

3A	To			
From	A	B	C	Supply
1	6	8	10	150
2	7	11	11	
3	4	5	12	275
Demand	200	100	300	



THE MINIMUM CELL COST SOLUTION

3B	To			
From	A	B	C	Supply
1	6	8	10	150
2	7	11	11	175
3	200 4	5	12	275 → 75
Demand	200 → 0	100	300	



THE MINIMUM CELL COST SOLUTION cont'd

3C	To			
From	A	B	C	Supply
1	6	8	10	150
2	7	11	11	175
3	4	5	12	75 → 0
	200	75		
Demand	0	100 → 25	300	



THE MINIMUM CELL COST SOLUTION

3D	To			
From	A	B	C	Supply
1	6	8	10	150 → 125
		25		
2	7	11	11	175
3	4	5	12	0
	200	75		
Demand	0	25 → 0	300	



THE MINIMUM CELL COST SOLUTION cont'd

3E	To						
From	A		B		C		Supply
1		6		8		10	125 → 0
			25		125		
2		7		11		11	175
3		4		5		12	0
	200		75				
Demand	0		0		300 → 175		



THE MINIMUM CELL COST SOLUTION cont'd

3F	To						
From	A		B		C		Supply
1		6		8		10	0
			25		125		
2		7		11		11	175 → 0
					175		
3		4		5		12	0
	200		75				
Demand	0		0		175 → 0		



THE MINIMUM CELL COST SOLUTION

3G	To						
From	A		B		C		Supply
1		6		8		10	150
			25		125		
2		7		11		11	175
					175		
3		4		5		12	275
	200		75				
Demand	200		100		300		



Initial Feasible Solution: MINIMUM CELL COST

ALLOCATED CELLS

The non-zero allocations from the table are:

- ▶ $x_{1B} = 25, \quad x_{1C} = 125$
- ▶ $x_{2C} = 175$
- ▶ $x_{3A} = 200, \quad x_{3B} = 75$



Initial Feasible Solution: MINIMUM CELL COST

ALLOCATED CELLS

The non-zero allocations from the table are:

- ▶ $x_{1B} = 25, \quad x_{1C} = 125$
- ▶ $x_{2C} = 175$
- ▶ $x_{3A} = 200, \quad x_{3B} = 75$

TOTAL COST

The total cost Z is defined as the sum of the product of unit costs and allocations:

$$Z = 8(x_{1B}) + 10(x_{1C}) + 11(x_{2C}) + 4(x_{3A}) + 5(x_{3B})$$

$$Z = 8(25) + 10(125) + 11(175) + 4(200) + 5(75) = \$4550$$



THE VOGEL'S APPROXIMATION METHOD (VAM)

1. The Vogel's approximation method is also called VAM.
2. This method is based on the concept of penalty cost or regret.
3. If the decision maker incorrectly chooses from several courses of action (or alternative routes), a penalty may be suffered, and the decision maker may regret the decision that was made.
4. In a transportation model, a wrong decision is allocating to a cell that contains not the lowest cost.
5. A penalty cost is the difference between the least and next least cell cost in a row or column.



THE VOGEL'S APPROXIMATION METHOD (VAM)

1. To begin the Vogel's approximation method,
 - 1.1 Determine the penalty cost for each row and column by subtracting the lowest cell cost in the row or column from the next lowest cell cost in the same row or column.
 - 1.2 Select the row or column with the highest penalty cost (breaking ties arbitrarily or choosing the lowest-cost cell).
 - 1.3 Allocate as much as possible to the feasible cell with the lowest transportation cost in the row or column with the highest penalty cost.
 - 1.4 Repeat steps 1,2, and 3 until all rim requirements have been satisfied.
2. The VAM solution for example 1 is shown below with the total cost in the bottom right corner of the tableau 4.



The VAM Solution.

4A	To				
From	A	B	C	Supply	Penalty
1	6	8	10	150	2
2	7	11	11	175 $\rightarrow$ 0	4
3	4	5	12	275	1
Demand	200 $\rightarrow$ 25	100	300		
Penalty	2	3	1		



The VAM Solution.

4B	To						
From	A		B		C	Supply	Penalty
1		6		8	10	150	2
2	175	7		11	11	0	
3		4		5	12	275 → 175	1
			100				
Demand	25	100 → 0		300			
Penalty	2	3		2			



The VAM Solution.

4C	To						
From	A		B		C	Supply	Penalty
1		6		8	10	150	4
2		7		11	11	0	
	175						
3		4		5	12	175 → 150	8
	25		100				
Demand	25 → 0		0		300		
Penalty	2				2		



The VAM Solution.

4D	To							
From	A		B		C	Supply	Penalty	
1		6		8		10	150	10
2		7		11		11	0	
	175							
3		4		5		12	150 → 0	12
	25		100		150			
Demand	0		0		300 → 150			
Penalty					2			



The VAM Solution.

4E	To						
From	A	B	C	Supply	Penalty		
1	6	8	10	150	150 $\rightarrow$ 0	10	
2	7	11	11				
	175			0			
3	4	5	12	150	0		
	25	100					
Demand	0	0	150 $\rightarrow$ 0				
Penalty			10				



The VAM Solution.

4F	To							
From	A		B		C	Supply	Penalty	
1		6		8		10	150	10
					150			
2		7		11		11	175	
	175							
3		4		5		12	275	
	25		100		150			
Demand	200		100		300			



ALLOCATED CELLS

The non-zero allocations from the VAM solution are:

- ▶ $x_{1C} = 150$
- ▶ $x_{2A} = 175$
- ▶ $x_{3A} = 25, \quad x_{3B} = 100, \quad x_{3C} = 150$

TOTAL COST

The total cost Z is calculated as the sum of (unit cost $\times$ allocation):

$$Z = 10(x_{1C}) + 7(x_{2A}) + 4(x_{3A}) + 5(x_{3B}) + 12(x_{3C})$$

$$Z = 10(150) + 7(175) + 4(25) + 5(100) + 12(150) = \$5125$$



Summary

- ▶ From the solutions [2], [3], and [4] we can observe that the least cost method converges faster than the north-west corner method and the VAM for the above example 1.
- ▶ Any of the initial solutions can be used to attain the optimal solution.
- ▶ We shall continue to optima using any of the following methods:
 1. Stepping-stone method
 2. Modified Distribution (MODI) method.



Algorithm 1: Demand-Based Allocation Method (DBAM)

- 1: Step 1: Select the column with the smallest remaining demand.
- 2: if there is a tie between columns then
- 3: choose the column with the smallest minimum cost
- 4: if still tied, choose the column with the smallest average cost
- 5: end if
- 6: Step 2: In the selected column, choose the row with the smallest cost among rows with remaining supply.
- 7: Step 3: Allocate the maximum feasible amount to the selected cell, equal to the smaller of the row's remaining supply and the column's remaining demand. Record the allocation in the tableau.
- 8: Step 4: Reduce the corresponding row supply and column demand by the allocated amount.



Algorithm 1: Demand-Based Allocation Method (DBAM) cont'd

- 9: Step 5: Determine the next pivot.
- 10: if both the row and column are exhausted then
- 11: return to Step 1
- 12: else if only the row is exhausted then
- 13: remain in the same column and return to Step 2
- 14: else if only the column is exhausted then
- 15: remain in the same row and select the next column with the smallest cost
- 16: end if
- 17: Repeat Steps 3–5 until all supplies, intermediary flows, and demands are satisfied.



THE INITIAL TRANSPORTATION SIMPLEX TABLEAU

1	To			
From	A	B	C	Supply
1	6	8	10	150
2	7	11	11	
3	4	5	12	275
Demand	200	100	300	



The DBAM Solution

	To			
From	A	B	C	Supply
1	6	8	10	150
2	7	11	11	175
3	4	5	12	275 → 175
Demand	200	100 → 0	300	



The DBAM Solution

	To			
From	A	B	C	Supply
1	6	8	10	150
2	7	11	11	175
3	4	5	12	175 → 0
	175	100		
Demand	200 → 25	0	300	



The DBAM Solution

	To			
From	A	B	C	Supply
1	6	8	10	150 → 125
	25			
2	7	11	11	175
3	4	5	12	0
	175	100		
Demand	25 → 0	0	300	



The DBAM Solution

	To						
From	A		B		C		Supply
1	25	6		8	125	10	125 → 0
2		7		11		11	175
3	175	4	100	5		12	0
Demand	0		0		300 → 175		



The DBAM Solution

	To					
From	A		B		C	Supply
1	25	6		8	10	0
					125	
2		7		11	11	175 → 0
					175	
3	175	4	100	5	12	0
Demand	0		0		175 → 0	



The DBAM Solution

	To			
From	A	B	C	Supply
1	6	8	10	150
	25		125	
2	7	11	11	175
			175	
3	4	5	12	275
	175	100		
Demand	200	100	300	



ALLOCATED CELLS

The non-zero allocations from the DBAM solution are:

- ▶ $x_{1A} = 25, \quad x_{1C} = 125$
- ▶ $x_{2C} = 175$
- ▶ $x_{3A} = 175, \quad x_{3B} = 100$

TOTAL COST

The total cost Z is calculated as the sum of (unit cost $\times$ allocation):

$$Z = 6(x_{1A}) + 10(x_{1C}) + 11(x_{2C}) + 4(x_{3A}) + 5(x_{3B})$$

$$Z = 6(25) + 10(125) + 11(175) + 4(175) + 5(100) = \$4525$$



Methods for Finding Optimal Solutions To Transportation Problems

The two (2) methods available for finding optimal solutions to transportation problems are:

- ▶ Stepping-Stone Method
- ▶ Modified Distribution Method (MODI)



THE STEPPING-STONE METHOD

- ▶ Once an initial solution is attained, the problem must be solved using either the stepping-stone method or MODI.
- ▶ The stepping-stone method determines whether or not, there exists an empty cell that would reduce cost when used.
- ▶ An empty cell that will reduce cost is called a potential entering variable.
- ▶ To evaluate the cost reduction potential of an empty cell, a closed path connecting used cells to the empty cell is identified.
- ▶ After all empty cells are evaluated, the one with the greatest cost reduction potential is the entering variable.



THE STEPPING-STONE METHOD (cont'd)

- ▶ When reallocating units to the entering variable, a minimum amount is subtracted on the stepping-stone path.
- ▶ The stepping-stone process is repeated until no empty cell reduces cost (optimum).
- ▶ A multiple optimal solution occurs when an empty cell has a cost change of zero and all other empty cells have positive or zero cost change.
- ▶ An alternate optimal solution is attained by allocating to the empty cell with a zero cost change.



THE STEPPING-STONE METHOD (cont'd)

- ▶ The steps of the stepping-stone method are summarized below.
 1. Determine the stepping-stone path and cost changes for each empty cell in the tableau.
 2. Allocate as much as possible to the empty cell with the greatest net decrease in cost.
 3. Repeat steps 1 and 2 until all empty cells have cost changes that indicate existence of an optimal solution.



THE STEPPING-STONE METHOD (cont'd)

- ▶ However, if we add (or subtract) one ton to an empty cell on a row (or column), we must subtract (or add) one ton from another allocation on the same row (or column) so as to satisfy all rim requirements.
- ▶ Note that when one unit is added to (or subtracted from) a cell, total cost is increased (or decreased) by a unit cost of that cell.
- ▶ Consider the least cost solution below.



THE STEPPING-STONE METHOD

X_{1A}	To					
From	A		B		C	Supply
1	-1	6		8	10	150
		$\rightarrow +$	25	$\downarrow -$	125	
2		7		11	11	175
					175	
3		4		5	12	275
	200	$\uparrow -$	75	$\leftarrow +$		
Demand	200		100		300	



THE STEPPING-STONE METHOD (cont'd)

$$x_{1A} \rightarrow x_{1B} \rightarrow x_{3B} \rightarrow x_{3A} \implies 6 - 8 + 5 - 4 = \$ \boxed{-1} \quad (1)$$



THE STEPPING-STONE METHOD

The stepping-stone path for cell X_{2A} is shown below.

X_{2A}	To						
From	A		B		C		Supply
1		6		8		10	150
			25	↓ −	125	← +	
2	−1	7		11		11	175
		→ +			175	↑ −	
3		4		5		12	275
	200	↑ −	75	← +			
Demand	200		100		300		



THE STEPPING-STONE METHOD (cont'd)

► From tableau 5,

$$x_{2A} \rightarrow x_{2C} \rightarrow x_{1C} \rightarrow x_{1B} \rightarrow x_{3B} \rightarrow x_{3A}$$

$$\Rightarrow 7 - 11 + 10 - 8 + 5 - 4 = \$ \boxed{-1} \quad (2)$$



THE STEPPING-STONE METHOD

The stepping-stone path for cell X_{2B} is shown below.

X_{2B}	To						
From	A		B		C		Supply
1	6		8		10		150
			25	↓ −	125	← +	
2	7		+2	11	11		175
				→ +	175	↑ −	
3	4		5		12		275
	200		75				
Demand	200		100		300		



THE STEPPING-STONE METHOD (cont'd)

- ▶ The stepping-stone path for cell X_{2B} is shown in the equation below.

$$x_{2B} \rightarrow x_{2C} \rightarrow x_{1C} \rightarrow x_{1B} \implies 11 - 11 + 10 - 8 = \$ \boxed{+2} \quad (3)$$



THE STEPPING-STONE METHOD

The stepping-stone path for X_{3C} is shown below.

X_{3C}	To						
From	A		B		C		Supply
1		6	25	8	125	10	150
				↓ +		← -	
2		7		11	175	11	175
3	200	4	75	5	+5	12	275
				→ -		↑ +	
Demand	200			100		300	



THE STEPPING-STONE METHOD (cont'd)

- ▶ The stepping-stone path for X_{3C} is shown in the equation below.

$$x_{3C} \rightarrow x_{1C} \rightarrow x_{1B} \longrightarrow x_{3B} \implies 12 - 10 + 8 - 5 = \$ \boxed{+5} \quad (4)$$



THE STEPPING-STONE METHOD (cont'd)

- ▶ Since the cells x_{1A} and x_{2A} have the same cost change of -1 , the above solution is not optimal.
- ▶ We shall select x_{1A} arbitrarily to enter the solution, and we shall supply a minimum of 25 units from its path.



STEPPING-STONE METHOD - ITERATION 1

- The stepping-stone path for X_{1B} is shown in the equation below.

X_{1B}	To						
From	A		B		C		Supply
1	25	6	+1	8	125	10	150
		→ +		↓ −			
2		7		11	175	11	175
3	175	4	100	5		12	275
		↑ −		← +			
Demand	200		100		300		



STEPPING-STONE METHOD - ITERATION 2

- The stepping-stone path for X_{1B} continuation is shown in the equation below.

X_{1B}	To					
From	A		B		C	Supply
1	25	6	+1	8	10	150
		→ −		↓ +	125	
2		7		11	11	175
					175	
3	175	4	100	5	12	275
		↑ +			← −	
Demand	200		100		300	



THE STEPPING-STONE METHOD (cont'd)

1. From the tableau above,

$$x_{1B} \rightarrow x_{3B} \rightarrow x_{3A} \rightarrow x_{1A} \implies 8 - 5 + 4 - 6 = \$ \boxed{+1} \quad (5)$$



THE STEPPING-STONE METHOD

X_{2A}	To					
From	A		B		C	Supply
1	25	6	$\downarrow -$	8	10	150
					125 +	
2	0	7		11	11	175
		+			175 -	
3	175	4	100	5	12	275
		$\uparrow -$		$\leftarrow +$		
Demand	200		100		300	



THE STEPPING-STONE METHOD (cont'd)

1. From the tableau above,

$$x_{2A} \rightarrow x_{1A} \rightarrow x_{1C} \rightarrow x_{2C} \implies 7 - 6 + 10 - 11 = \$\boxed{0} \quad (6)$$



THE STEPPING-STONE METHOD

X_{2B}	To						
From	A		B		C		Supply
1	25	6		8	125	10	150
		↓ −				← +	
2		7	+3	11	175	11	175
				+		→ −	
3	175	4	100	5		12	275
		→ +		↑ −			
Demand	200		100		300		



THE STEPPING-STONE METHOD (cont'd)

1. From the tableau above,

$$\begin{aligned} x_{2B} &\rightarrow x_{3B} \rightarrow x_{3A} \rightarrow x_{1A} \rightarrow x_{1C} \rightarrow x_{2C} \\ \implies 11 - 5 + 4 - 6 + 10 - 11 &= \$ \boxed{+3} \end{aligned} \quad (7)$$



THE STEPPING-STONE METHOD

X_{3C}	To			
From	A	B	C	Supply
1	6 25 → +	8	10 125 ↓ -	150
2	7	11	11 175	175
3	4 175 ↑ -	5 100	$+4$ 12 ← +	275
Demand	200	100	300	



THE STEPPING-STONE METHOD (cont'd)

1. From the tableau above

$$x_{3C} \rightarrow x_{3A} \rightarrow x_{1A} \rightarrow x_{1C} \implies 12 - 4 + 6 - 10 = \$ \boxed{+4} \quad (8)$$



THE STEPPING-STONE METHOD (end)

1. Since all the cells have a positive cost change, an optimum is reached.
2. The optimal solution is:
 - ▶ $x_{1A} = 25$, $x_{1C} = 125$, $x_{2C} = 175$, $x_{3A} = 175$, $x_{3B} = 100$
3. The minimum total cost Z is calculated as follows:

$$Z = 6(x_{1A}) + 10(x_{1C}) + 11(x_{2C}) + 4(x_{3A}) + 5(x_{3B})$$

$$Z = 6(25) + 10(125) + 11(175) + 4(175) + 5(100) = \$4525$$

4. There exists an alternate solution since the cell x_{2A} has a zero cost change.
5. This means that if x_{2A} enters the basis, the current optimal solution will not be improved.
6. Let us move to the next slide for the MODI.



MODIFIED DISTRIBUTION (MODI) METHOD

1. MODI is a modified version of the stepping-stone method in which mathematical equations

$$u_i + v_j = c_{ij} \quad (9)$$

replace the stepping-stone paths.

2. $[u_i + v_j]$ is the intersection of source i and destination j .
3. Each MODI allocation replicates the stepping-stone allocation.
4. After each allocation to an empty cell, the u_i and v_j must be recomputed.



MODIFIED DISTRIBUTION (MODI) METHOD

1. The steps of the MODI method are:
 - 1.1 Develop an initial feasible solution using one of the three methods available.
 - 1.2 Solve for u_i and v_j such that $u_i + v_j = c_{ij}$.
 - 1.3 Compute the cost change k_{ij} for each empty cell using $k_{ij} = c_{ij} - u_i - v_j$.
 - 1.4 Allocate as much as possible to the empty cell that will result in the greatest net decrease in cost (most negative k_{ij}). Allocate according to the stepping-stone path for the selected cell.
 - 1.5 Repeat steps 2 through 4 until all k_{ij} are positive or zero.
2. Now, consider the least cost solution 7 below.



MODI (Iteration 1)

7	To						
From	A		B		C		Supply
1		6	25	8	125	10	150
2		7		11	175	11	175
3	200	4	75	5		12	275
Demand	200		100		300		



MODI EQUATIONS AND

1. From tableau 7:

$$u_1 = 0 \quad (10)$$

$$u_1 + v_B = 8 \quad (11)$$

$$u_1 + v_C = 10 \quad (12)$$

$$u_2 + v_C = 11 \quad (13)$$

$$u_3 + v_A = 4 \quad (14)$$

$$u_3 + v_B = 5 \quad (15)$$

SOLUTION

1. Equation (10) $\rightarrow$ (11, 12) :

$$v_B = 8, \quad v_C = 10 \quad (16)$$

1. Equation (16) $\rightarrow$ (13, 15) :

$$u_2 = 1, \quad u_3 = -3 \quad (17)$$

1. Equation (17) $\rightarrow$ (14) :

$$v_A = 7 \quad (18)$$



COST CHANGE (k_{ij})

$$k_{ij} = c_{ij} - u_i - v_j \quad (19)$$

$$\begin{aligned} k_{1A} &= c_{1A} - u_1 - v_A \\ \Rightarrow k_{1A} &= 6 - 0 - 7 = \$ \boxed{-1} \quad (20) \end{aligned}$$

$$\begin{aligned} k_{2A} &= c_{2A} - u_2 - v_A \\ k_{2A} &= 7 - 1 - 7 = \$ \boxed{-1} \quad (21) \end{aligned}$$

Cont'd

$$\begin{aligned} k_{2B} &= c_{2B} - u_2 - v_B \\ k_{2B} &= 11 - 1 - 8 = \$ \boxed{+2} \quad (22) \end{aligned}$$

$$\begin{aligned} k_{3C} &= c_{3C} - u_3 - v_C \\ k_{1A} &= 12 - (-3) - 10 = \$ \boxed{+5} \quad (23) \end{aligned}$$

1. The current solution is not optimal since $k_{1A} = k_{2A} < 0$.



1. A stepping-stone path determined for x_{1A} is shown on the next slide.



MODIFIED DISTRIBUTION (MODI)

8	To					
From	A		B		C	Supply
1	-1	6		8	10	150
		→ +	25	↓ -	125	
2		7		11	11	175
					175	
3		4		5	12	275
	200	↑ -	75	← +		
Demand	200		100		300	



1. Make a minimum allocation along the stepping-stone path for X_{1A} as shown below, to attain a new basic feasible solution.



MODI (End of Iteration 1)

9	To						
From	A		B		C		Supply
1	25	6		8	125	10	150
		→ +		↓ −			
2		7		11	175	11	175
3	175	4	100	5		12	275
		↑ −		← +			
Demand	200		100		300		



1. Develop MODI equations and solve them.
2. Evaluate the cost reduction potential for empty cells and determine the next entering variable.
3. See the next slide.



MODI EQUATIONS AND

$$u_1 = 0 \quad (24)$$

$$u_1 + v_A = 6 \quad (25)$$

$$u_1 + v_C = 10 \quad (26)$$

$$u_2 + v_C = 11 \quad (27)$$

$$u_3 + v_A = 4 \quad (28)$$

$$u_3 + v_B = 5 \quad (29)$$

SOLUTION

1. Equation (24) $\rightarrow$ (25, 26) :

$$v_A = 6, \quad v_C = 10 \quad (30)$$

1. Equation (30) $\rightarrow$ (27, 28) :

$$u_2 = 1, \quad u_3 = -2 \quad (31)$$

1. Equation (31) $\rightarrow$ (29) :

$$v_B = 7 \quad (32)$$



COST CHANGE (k_{ij})

1. Given that

$$k_{ij} = c_{ij} - u_i - v_j \quad (33)$$

$$k_{1B} = c_{1B} - u_1 - v_B$$

$$\Rightarrow k_{1B} = 8 - 0 - 7 = \$ \boxed{+1} \quad (34)$$

$$k_{2A} = c_{2A} - u_2 - v_A$$

$$k_{2A} = 7 - 1 - 6 = \$ \boxed{0} \quad (35)$$

Cont'd

$$k_{2B} = c_{2B} - u_2 - v_B$$

$$k_{2B} = 11 - 1 - 8 = \$ \boxed{+3} \quad (36)$$

$$k_{3C} = c_{3C} - u_3 - v_C$$

$$k_{3C} = 12 - (-2) - 10 = \$ \boxed{+4} \quad (37)$$

1. The current solution shown above is optimal since all $k_{ij} \geq 0$.



MODI Method: Optimal Solution

The optimal solution is:

$$\blacktriangleright x_{1A} = 25, \quad x_{1C} = 125, \quad x_{2C} = 175, \quad x_{3A} = 175, \quad x_{3B} = 100$$

The minimum total cost Z is calculated as follows:

$$Z = 6(x_{1A}) + 10(x_{1C}) + 11(x_{2C}) + 4(x_{3A}) + 5(x_{3B})$$

$$Z = 6(25) + 10(125) + 11(175) + 4(175) + 5(100) = \$4525$$

Remarks:

1. Multiple optimal solutions exist since multiple zeros indicate multiple optimal solutions.



The Network Solution

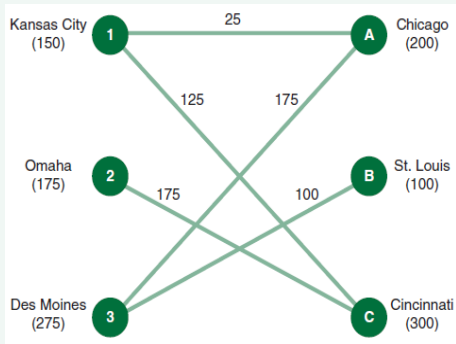


Figure: The Transportation Network Solution.



Exercises



Exercise 1

1. Green Valley Mills produces carpet at plants in St. Louis and Richmond. The carpet is then shipped to two outlets, located in Chicago and Atlanta. The cost per ton of shipping carpet from each of the two plants to the two warehouses is as follows:

From	To (Cost)	
	A. Chicago	Atlanta
1. St. Louis	\$40	\$65
2. Richmond	70	30

2. The plants at St. Louis and Richmond can supply 250 tons and 400 tons of carpet per week, respectively. The Chicago outlet has a demand of 300 tons per week, and the outlet at Atlanta demands 350 tons per week. The company wants to know the number of tons of carpet to ship from each plant to each outlet so as to minimize total shipping cost.
3. Solve this transportation problem.



Exercise 2

- Below is a transportation problem with costs, supply, and demand:

	To			
From	1	2	3	Supply
A	6	7	4	100
B	5	3	6	180
C	8	5	7	200
Demand	135	175	170	

- Find the optimal solution, using the stepping-stone or the MODI method. Use the least cost method for the starting solution.



THE TRANSSHIPMENT MODEL

Introduction

1. The transshipment model is an extension of the transportation model in which intermediate transshipment points are added between the sources and destinations.
2. An example of a transshipment point is a distribution center or warehouse located between plants and stores.
3. In a transshipment problem, items may be transported from sources through transshipment points on to destinations, from one source to another, from one transshipment point to another, from one destination to another, or directly from sources to destinations, or some combination of these alternatives.



THE TRANSSHIPMENT MODEL

EXAMPLE

We will expand our wheat-shipping example to demonstrate the formulation of a transshipment model.

1. Wheat is harvested at farms in Nebraska and Colorado before being shipped to the points. The amount of wheat harvested at each farm is 300 tons. The wheat is then shipped to the mills in Chicago, St. Louis, and Cincinnati. The shipping costs from the grain elevators to the mills remain the same, and the shipping costs from the farms to the grain elevators are as follows:



The Transshipment Model

Cont.

	Grain Elevator		
Farm	3. Kansas City	4. Omaha	5. Des Moines
1. Nebraska	\$16	\$10	\$12
2. Colorado	15	14	17

1. The basic structure of this model is shown in the figure below.



The Transshipment Model

Cont'd

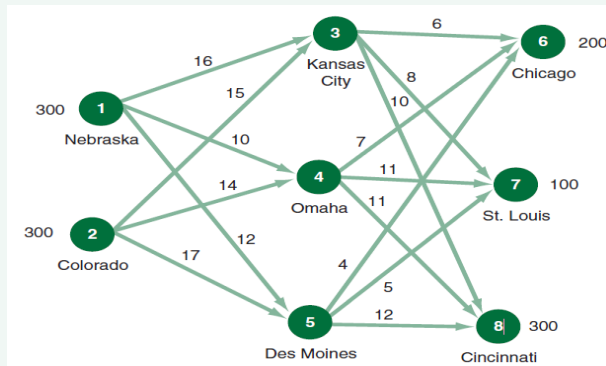


Figure: Network of transshipment routes.



The Transshipment Model

Cont'd

1. As with the transportation problem, a linear programming model is developed with supply and demand constraints.
2. The available supply constraints for the farms in Nebraska and Colorado are

▶ $x_{13} + x_{14} + x_{15} = 300$

▶ $x_{23} + x_{24} + x_{25} = 300$



The Transshipment Model

Cont'd

1. The demand constraints at the Chicago, St. Louis, and Cincinnati mills are

▶ $x_{36} + x_{46} + x_{56} = 200$

▶ $x_{37} + x_{47} + x_{57} = 100$

▶ $x_{38} + x_{48} + x_{58} = 300$



The Transshipment Model

Cont'd

1. At each trans-shipment point, whatever is shipped in must also be shipped out.
2. Thus, the following constraints are developed for the trans-shipment points.
 - ▶ $x_{13} + x_{23} - x_{36} - x_{37} - x_{38} = 0$
 - ▶ $x_{14} + x_{24} - x_{46} - x_{47} - x_{48} = 0$
 - ▶ $x_{15} + x_{25} - x_{56} - x_{57} - x_{58} = 0$
3. The complete linear programming model, including the objective function, is summarized as follows:



The Transshipment Model

cont'd

Min.

$$Z = 16x_{13} + 10x_{14} + 12x_{15} + 15x_{23} + 14x_{24}$$

$$+ 17x_{25} + 6x_{36} + 8x_{37} + 10x_{38} + 7x_{46}$$

Objective ↑

$$+ 11x_{47} + 11x_{48} + 4x_{56} + 5x_{57} + 12x_{58}$$

S.t.

$$x_{13} + x_{14} + x_{15} = 300$$

Supply ↑

$$x_{23} + x_{24} + x_{25} = 300$$

$$x_{36} + x_{46} + x_{56} = 200$$

$$x_{37} + x_{47} + x_{57} = 100$$

Demand ↑

$$x_{38} + x_{48} + x_{58} = 300$$

$$x_{13} + x_{23} - x_{36} - x_{37} - x_{38} = 0$$

$$x_{14} + x_{24} - x_{46} - x_{47} - x_{48} = 0$$

$$x_{15} + x_{25} - x_{56} - x_{57} - x_{58} = 0$$

Transshipment ↑

Non-negativity →

$$x_{ij} \geq 0$$



The Trans-shipment Solution process.

1. We can find an optimal solution to the transshipment problem by solving a transportation problem.
2. The steps are shown below:

Note the following:

1. Shipments to the dummy and from a point to itself will have a zero shipping cost.
2. Supply and demand at all transshipment points equal the total supplies at all sources.
3. Each supply point will have a supply equal to its original supply, and each demand point will have a demand equal to its original demand.
4. Balance the problem where necessary.



The Trans-shipment Solution process.

1. The following table shows the per unit cost between nodes:

Sources	Destinations					
	3	4	5	6	7	8
1	16	10	12	16	17	21
2	15	14	17	21	22	25
3	0	13	10	6	8	10
4	13	0	11	7	11	11
5	10	11	0	4	5	12

1. These values were obtained from the network.
2. 16, 17, 21, 21, 22 and 25 are the minimum costs for X_{16} , X_{17} , X_{18} , X_{26} , X_{27} and X_{28} , respectively, choosing the best alternatives.
3. Below is the initial simplex tableau for the problem.



The Initial Transportation Tableau

	To						
From	3	4	5	6	7	8	Supply
1	16	10	12	16	17	21	300
2	15	14	17	21	22	25	300
3	0	13	10	6	8	10	600
4	13	0	11	7	11	11	600
5	10	11	0	4	5	12	600
Demand	600	600	600	200	100	300	2400



Trans-shipment Solution process.

Solution cont'd

1. The next tableau shows the least cost solution to the problem.



The Least Cost Solution

	To						
From	3	4	5	6	7	8	Supply
1	16	10	12	16	17	21	300
2	15	14	17	21	22	25	
3	0	13	10	6	8	10	600
4	13	0	11	7	11	11	
5	10	11	0	4	5	12	600
Demand	600	600	600	200	100	300	\$12,400



Trans-shipment Solution process.

Solution cont'd

1. Since degeneracy is seen in tableau 12, we must make four empty cells basic variables with zero allocations.
2. This result is shown in tableau 13 from which we shall derive the MODI equations.



The Least Cost Solution

13 From	To						
	3	4	5	6	7	8	Supply
1	16	10	12	16	17	21	300
			0	200	100		
2	15	14	17	21	22	25	300
						300	
3	0	13	10	6	8	10	600
	600			0		0	
4	13	0	11	7	11	11	600
		600		0			
5	10	11	0	4	5	12	600
			600				
Demand	600	600	600	200	100	300	\$12,400



The Transshipment Solution Process

THE MODI

$$u_1 = 0 \quad (38)$$

$$u_1 + v_5 = 12 \quad (39)$$

$$u_1 + v_6 = 16 \quad (40)$$

$$u_1 + v_7 = 17 \quad (41)$$

$$u_2 + v_8 = 25 \quad (42)$$

EQUATIONS

$$u_3 + v_3 = 0 \quad (43)$$

$$u_3 + v_6 = 6 \quad (44)$$

$$u_3 + v_8 = 10 \quad (45)$$

$$u_4 + v_4 = 0 \quad (46)$$

$$u_4 + v_6 = 7 \quad (47)$$

$$u_5 + v_5 = 0 \quad (48)$$



The Transshipment Solution Process

SOLUTION

1. Putting Equation

(38) $\rightarrow$ (39, 40, 41) :

$$v_5 = 12, \quad v_6 = 16 \quad (49)$$

$$v_7 = 17 \quad (50)$$

1. Putting Equation

(49, 50) $\rightarrow$ (44, 47, 48) :

$$u_3 = -10, \quad u_4 = -9 \quad (51)$$

$$u_5 = -12 \quad (52)$$

con't

1. Putting Equation

(51, 52) $\rightarrow$ (43, 45, 46) :

$$v_3 = 10, \quad v_4 = 9 \quad (53)$$

$$v_8 = 20 \quad (54)$$

1. Putting Equation (54) $\rightarrow$ (42) :

$$u_2 = 5 \quad (55)$$



The Transshipment Solution Process

COST CHANGE (k_{ij})

1. Given that

$$k_{ij} = c_{ij} - u_i - v_j \quad (56)$$

$$k_{13} = c_{13} - u_1 - v_3$$

$$k_{13} = 16 - 0 - 10 = \$ \boxed{+6} \quad (57)$$

$$k_{14} = c_{14} - u_1 - v_4$$

$$k_{14} = 10 - 0 - 9 = \$ \boxed{+1} \quad (58)$$

Cont'd

$$k_{23} = c_{23} - u_2 - v_3$$

$$k_{23} = 15 - 5 - 10 = \$ \boxed{0} \quad (59)$$

$$k_{24} = c_{24} - u_2 - v_4$$

$$k_{24} = 14 - 5 - 9 = \$ \boxed{0} \quad (60)$$

$$k_{25} = c_{25} - u_2 - v_5$$

$$k_{25} = 17 - 12 - 5 = \$ \boxed{0} \quad (61)$$



The Transshipment Solution Process

COST CHANGE (k_{ij})

$$k_{26} = c_{26} - u_2 - v_6$$

$$k_{26} = 21 - 5 - 16 = \$ \boxed{0} \quad (62)$$

$$k_{27} = c_{27} - u_2 - v_7$$

$$k_{27} = 22 - 5 - 17 = \$ \boxed{0} \quad (63)$$

$$k_{34} = c_{34} - u_3 - v_4$$

$$k_{34} = 13 + 10 - 9 = \$ \boxed{+14} \quad (64)$$

Cont'd

$$k_{35} = c_{35} - u_3 - v_5$$

$$k_{35} = 10 + 10 - 12 = \$ \boxed{+8} \quad (65)$$

$$k_{37} = c_{37} - u_3 - v_7$$

$$k_{37} = 8 + 10 - 17 = \$ \boxed{+1} \quad (66)$$

$$k_{43} = c_{43} - u_4 - v_3$$

$$k_{43} = 13 + 9 - 10 = \$ \boxed{+12} \quad (67)$$



The Transshipment Solution Process

COST CHANGE (k_{ij})

$$k_{45} = c_{45} - u_4 - v_5$$

$$k_{45} = 11 + 9 - 12 = \$ \boxed{+8} \quad (68)$$

$$k_{47} = c_{47} - u_4 - v_7$$

$$k_{47} = 11 + 9 - 17 = \$ \boxed{+3} \quad (69)$$

$$k_{48} = c_{48} - u_4 - v_8$$

$$k_{48} = 11 + 9 - 20 = \$ \boxed{0} \quad (70)$$

Cont'd

$$k_{53} = c_{53} - u_5 - v_3$$

$$k_{53} = 10 + 12 - 10 = \$ \boxed{+12} \quad (71)$$

$$k_{54} = c_{54} - u_5 - v_4$$

$$k_{54} = 11 + 12 - 9 = \$ \boxed{+14} \quad (72)$$

$$k_{56} = c_{56} - u_5 - v_6$$

$$k_{56} = 4 + 12 - 16 = \$ \boxed{0} \quad (73)$$



The Transshipment Solution Process

COST CHANGE (END)

$$k_{57} = c_{57} - u_5 - v_7$$

$$k_{57} = 5 + 12 - 17 = \$\boxed{0} \quad (74)$$

$$k_{58} = c_{58} - u_5 - v_8$$

$$k_{58} = 12 + 12 - 20 = \$\boxed{+4} \quad (75)$$

1. The current solution is optimal since all cost changes $k_{ij} \geq 0$.

Remarks:

1. Due to the degenerate condition in tableau $\boxed{12}$, more iterations may be done when tracing the optimal solution.
2. This depends on the choice of arbitrary basic variables from the degenerate tableau.
3. See the shipping list and optimal transportation network below.



The transshipment Solution Process

SHIPPING LIST

Source	Destination	Shipments	Shipping Cost
1	6	200	$200 \times 16 = \$3,200$
1	7	100	$100 \times 17 = \$1,700$
1	5	0	$0 \times 12 = 0$
2	8	300	$300 \times 25 = \$7,500$
3	3	600	$600 \times 0 = 0$
3	6	0	$0 \times 6 = 0$
3	8	0	$0 \times 10 = 0$
4	4	600	$600 \times 0 = 0$
4	6	0	$0 \times 7 = 0$
5	5	600	$600 \times 0 = 0$
Total Cost			\$12,400



The Network Solution

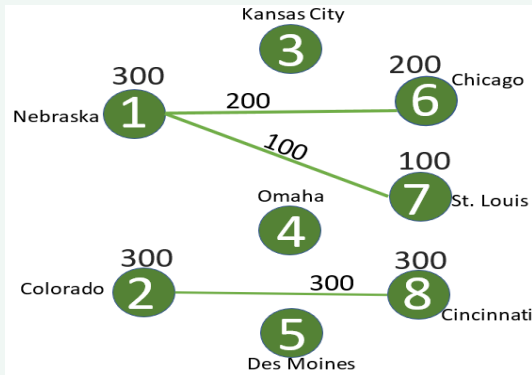


Figure: The Transshipment Network Solution.



THE ASSIGNMENT MODEL

Introduction

1. The assignment model is a special form of a linear programming model that is similar to the transportation model.
2. There are differences, however.
3. In the assignment model, the supply at each source and the demand at each destination are each limited to one unit.
4. The following example will demonstrate the assignment model.



The Assignment Model

Example

1. The Atlantic Coast Conference (ACC) has four basketball games on a particular night. The conference office wants to assign four teams of officials to the four games in a way that will minimize the total distance travelled by the officials. The supply is always one team of officials, and the demand is for only one team of officials at each game. The distances in miles for each team of officials to each game location are shown in the following table:



The Assignment Model

Cont'd

		Game Sites		
Officials	Raleigh	Atlanta	Dallas	Clemson
A	210	90	180	160
B	100	70	130	200
C	175	105	140	170
D	80	65	105	120

1. In the linear programming formulation of the assignment model, all the supply values for each source equal one, and all the demand values at each destination equal one.
2. Thus, the example is formulated as follows:



The Assignment Model

cont'd

$$\begin{aligned} \text{Min. } Z = & 210x_{AR} + 90x_{AA} + 180x_{AD} + 160x_{AC} + 100x_{BR} + \\ & 70x_{BA} + 130x_{BD} + 200x_{BC} + 175x_{CR} + 105x_{CA} + \\ & 140x_{CD} + 170x_{CC} + 80x_{DR} + 65x_{DA} + 105x_{DD} + 120x_{DC} \end{aligned}$$

$$\begin{aligned} \text{S.t. } & x_{AR} + x_{AA} + x_{AD} + x_{AC} = 1 \\ & x_{BR} + x_{BA} + x_{BD} + x_{BC} = 1 \\ & x_{CR} + x_{CA} + x_{CD} + x_{CC} = 1 \\ & x_{DR} + x_{DA} + x_{DD} + x_{DC} = 1 \\ & x_{AR} + x_{BR} + x_{CR} + x_{DR} = 1 \\ & x_{AA} + x_{BA} + x_{CA} + x_{DA} = 1 \\ & x_{AD} + x_{BD} + x_{CD} + x_{DD} = 1 \\ & x_{AC} + x_{BC} + x_{CC} + x_{DC} = 1 \\ & x_{ij} \geq 0 \end{aligned}$$



The Assignment Model

Summary

1. An assignment model is for a special form of transportation problem in which all supply and demand values equal one.
2. The above assignment model is balanced.
3. An unbalanced model exists when supply exceeds demand or demand exceeds supply.



1. There are several methods for solving the assignment model.
2. We shall use the Hungarian method for the above model example.
3. An opportunity cost table must be developed by performing row reductions and column reductions.
4. Assignments are made to locations with zeros in the opportunity cost table.
5. An optimal solution occurs when the number of independent unique assignments equal the number of rows or columns.
6. If the number of unique assignments is less than the number of rows (or columns), a line test must be used.



1. When supply exceeds demand, a dummy column is added to the assignment tableau.
2. When demand exceeds supply, a dummy row is added to the assignment tableau.
3. Assignments are made to locations with zeros in the opportunity cost table.
4. An optimal solution occurs when the number of independent unique assignments equal the number of rows or columns.
5. A prohibited assignment is given a relative cost of M so that it will never be selected.



Assignment solution steps

1. The steps of the assignment solution method are :
 - 1.1 Perform row reductions by subtracting the minimum value in each row from all row values.
 - 1.2 Perform column reductions by subtracting the minimum value in each column from all column values.
 - 1.3 In the completed opportunity cost table, cross out all zeros, using the minimum number of horizontal or vertical lines.
 - 1.4 If fewer than m lines are required, subtract the minimum uncrossed value from all other uncrossed values and add it to all cells where two lines intersect. Leave all other values unchanged, and repeat step 3.
 - 1.5 If m lines are required, m unique assignments can be made (optimum). Repeat step 4 if fewer than m lines are required.



Row reductions

		Game Sites		
Officials	Raleigh	Atlanta	Dallas	Clemson
A	120	0	90	70
B	30	0	60	130
C	70	0	35	65
D	15	0	40	55



Column reductions

Officials	Raleigh	Atlanta	Dallas	Clemson
A	105	0	55	15
B	15	0	25	75
C	55	0	0	10
D	0	0	5	0



Column reductions

Officials	Raleigh	Atlanta	Dallas	Clemson
A	105	0	55	15
B	15	0	25	75
C	55	0	0	10
D	0	0	5	0



Column reductions

Officials	Raleigh	Atlanta	Dallas	Clemson
A	105	0	55	15
B	15	0	25	75
C	55	0	0	10
D	0	0	5	0



Column reductions

Officials	Raleigh	Atlanta	Dallas	Clemson
A	105	0	55	15
B	15	0	25	75
C	55	0	0	10
D	0	0	5	0



Assignment solution steps

1. There are three unique assignments meanwhile, four are required to reach optima.
2. The next iteration is shown below.

		Game Sites		
Officials	Raleigh	Atlanta	Dallas	Clemson
A	90	0	40	0
B	0	0	10	60
C	55	15	0	10
D	0	15	5	0



Assignment solution steps

		Game Sites		
Officials	Raleigh	Atlanta	Dallas	Clemson
A	90	0	40	0
B	0	0	10	60
C	55	15	0	10
D	0	15	5	0



Assignment solution steps

	Game Sites			
Officials	Raleigh	Atlanta	Dallas	Clemson
A	90	0	40	0
B	0	0	10	60
C	55	15	0	10
D	0	15	5	0



Assignment solution steps

	Game Sites			
Officials	Raleigh	Atlanta	Dallas	Clemson
A	90	0	40	0
B	0	0	10	60
C	55	15	0	10
D	0	15	5	0



Assignment solution steps

	Game Sites			
Officials	Raleigh	Atlanta	Dallas	Clemson
A	90	0	40	0
B	0	0	10	60
C	55	15	0	10
D	0	15	5	0



Assignment solution steps

	Game Sites			
Officials	Raleigh	Atlanta	Dallas	Clemson
A	90	0	40	0
B	0	0	10	60
C	55	15	0	10
D	0	15	5	0



Assignment solution steps

	Game Sites			
Officials	Raleigh	Atlanta	Dallas	Clemson
A	90	0	40	0
B	0	0	10	60
C	55	15	0	10
D	0	15	5	0



Assignment solution steps

Officials	Game Sites			
	Raleigh	Atlanta	Dallas	Clemson
A	90	0	40	0
B	0	0	10	60
C	55	15	0	10
D	0	15	5	0



Assignment solution steps

1. Since at least four lines are required to cross out all the zeros, this indicates that four unique assignments can be made and that an optimal solution has been reached.
2. Now, let us make the assignments from the above table.
3. Remember that assignments can be made to cells containing zeros and once an assignment is made to a row or column, no assignment can be made to that row or column.
4. The assignments and their respective distances are summarized on the next slide.



Assignment Solution Methods

ASSIGNMENT A

Official	Site	Distance
A	Atlanta	90
B	Raleigh	100
C	Dallas	140
D	Clemson	120
Total		450 miles

ASSIGNMENT B

Official	Site	Distance
A	Clemson	160
B	Atlanta	70
C	Dallas	140
D	Raleigh	80
Total		450 miles

Remarks:

1. Multiple solutions exist for the above problem since multiple zeros indicate multiple solutions.



Exercises



Exercise 1

1. A shop has four machinists to be assigned to four machines. The hourly cost of having each machine operated by each machinist is as follows:

Machinist	Machine (Cost/hr.)			
	A	B	C	D
1	\$12	\$11	\$8	\$14
2	10	9	10	8
3	14	8	7	11
4	6	8	10	9

However, because he does not have enough experience, machinist 3 cannot operate machine B.

- Determine the optimal assignment and compute total minimum cost.
- Formulate this problem as a general linear programming model.



Exercise 2

1. The Omega pharmaceutical firm has five salespersons, whom the firm wants to assign to five sales regions. Given their various previous contacts, the salespersons are able to cover the regions in different amounts of time. The amount of time (days) required by each salesperson to cover each city is shown in the following table:

Salesperson	Region (days)				
	A	B	C	D	E
1	17	10	15	16	20
2	12	9	16	9	14
3	11	16	14	15	12
4	14	10	10	18	17
5	13	12	9	15	11

Which salesperson should be assigned to each region to minimize total time? Compute the total minimum time for the optimal assignments.



Exercise 3

1. The Bunker Manufacturing firm has five employees and six machines and wants to assign the employees to the machines to minimize cost. A cost table showing the cost incurred by each employee on each machine follows:

Employee	Machine					
	A	B	C	D	E	F
1	\$12	\$7	\$20	\$14	\$8	\$10
2	10	14	13	20	9	11
3	5	3	6	9	7	10
4	9	11	7	16	9	10
5	10	6	14	8	10	12

Because of union rules regarding departmental transfers, employee 3 and 4 cannot be assigned to machine E and B respectively.

2. Indicate the optimal assignments and compute total minimum cost.



Exercise 4

1. Given the following cost table for an assignment problem, determine the optimal assignment and compute total minimum cost:

Operator	Machine			
	A	B	C	D
1	\$10	\$2	\$8	\$6
2	9	5	11	9
3	12	7	14	14
4	3	1	4	2

Show an alternate solution if there are multiple optima.



Exercise 5

1. An electronics firm produces electronic components, which it supplies to various electrical manufacturers. Quality control records indicate that different employees produce different numbers of defective items. The average number of defects produced by each employee for each of six components is given in the following table:

Employee	Component					
	A	B	C	D	E	F
1	30	21	16	26	30	22
2	22	28	14	30	20	13
3	18	16	25	14	12	22
4	14	22	18	23	21	30
5	25	18	14	16	116	28
6	32	14	10	14	18	20

Determine the optimal assignment that will minimize the total average number of defects produced by the firm per month.



Exercise 6

1. A dispatcher for Citywide Taxi Company has six taxi cabs at different locations and five customers who have called for service. The mileage from each taxi's present location to each customer is shown in the following table:

Cab	Customer				
	1	2	3	4	5
A	7	2	4	10	7
B	5	1	5	6	6
C	8	7	6	5	5
D	2	5	2	4	5
E	3	3	5	8	4
F	6	2	4	3	4

Determine the optimal assignment(s) that will minimize the total mileage travelled.



Exercise 7

1. A university department head has five instructors to be assigned to four different courses. All the instructors have taught the courses in the past and have been evaluated by the students. The rating for each instructor for each course is given in the following table (a perfect score is 100):

Instructor	Course			
	A	B	C	D
1	80	75	90	85
2	95	90	90	97
3	85	95	88	91
4	93	91	80	84
5	91	92	93	88

The department head wants to assign instructors to courses to maximize overall average evaluation. The instructor who is not assigned to teach a course will be assigned to grade exams.



Exercise 8

1. Carolina Airlines, a small commuter airline in North Carolina, has six flight attendants that it wants to assign to six monthly flight schedules in a way that will minimize the number of nights they will be away from their homes. The numbers of nights each attendant must be away from home with each schedule are given in the following table:

Attendant	Schedule					
	A	B	C	D	E	F
1	7	4	6	10	5	8
2	4	5	5	12	7	6
3	9	9	11	7	10	8
4	11	6	8	5	9	10
5	5	8	6	10	7	6
6	10	12	11	9	9	10

Identify the optimal assignments that will minimize the total number of nights the attendants will be away from home.



Exercise 9

1. The Vanguard Publishing Company wants to hire seven of the eight college students who have applied as salespeople to sell encyclopedias during the summer. The company desires to allocate them to three sales territories. Territory 1 requires three salespeople, and territories 2 and 3 require two salespeople each. The following table shows the amounts of sales per day:

Salesperson	Territory		
	1	2	3
A	\$110	\$150	\$130
B	90	120	80
C	205	160	175
D	125	100	115
E	140	105	150
F	100	140	120
G	180	210	160
H	110	120	90

Help the company allocate the salespeople to maximize sales.





J. K. Sharma (2010)

Quantitative Methods: Theory and Applications

ISBN 10: 0230-32871-7



J. K. Sharma (2016)

Operations Research: Theory and Applications, 6th Edition

ISBN: 978-93-85935-14-5



B. W. Taylor (2013)

Introduction to management science, 11th Edition

ISBN-13: 978-0-13-275191-9

ISBN-10: 0-13-275191-7.

